Chart-Aware Contiguous Memory for Discretized Optical Phase Space: Geometry, Addressing, and the Limits of a Four-Day Prototype

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Abstract

We develop a chart-aware, contiguous-memory architecture for discretized optical phase space, separating ray dynamics, measurement observables, geometric discretization, and address correctness. A progressively ellipsoidal optical domain motivates the use of Hamiltonian characteristics and local phase-space charts rather than conventional Cartesian indexing. We derive the geometric conditions under which the domain is well posed, show that spatially indexed memory loses ray-branch identity at caustics, and formulate the address construction as a multi-chart encoder with explicit range and injectivity invariants, released as a verification-ready Ada/SPARK specification with a GNATprove transcript (47/47 proof obligations discharged, 0 unproved, for this encoder instance and its contract structure). Under a fixed hardware budget we formalize cell allocation as a task-aware quantization problem via the law of total variance, with calibration-effort costs estimated from finite samples. On a released calibrated synthetic model, a chart-legal dynamic-programming partition improves held-out timing distortion by $\mathbf{1.75}\times$ over uniform angular binning (4.230×10^{-4} vs 2.423×10^{-4}; ratio 1.746 ± 0.005 at 95% CI, 30 sampling seeds). We report controls that bound what this number means: the gain falls from $2.41\times$ to $1.51\times$ as the budget grows from $K = 16$ to $K = 256$; a single smooth delay law with no two-regime structure already gives $1.50\times$; and because the delay law is known in closed form, analytic quadrature matches the sample-estimated partition to 0.07%. We further price the chart-legal constraint by computing what it forbids: optimizing the scalar observable alone reaches $10.6\times$ lower distortion, but its cells fragment into 91 angular runs against 32, so it is not representable at equal address cost. Chart legality is therefore an address-structure cost, not a moral one, and topology, identity, address count, calibration, and distortion are coupled resources. We unify these results under a single criterion, *observable-sufficient representation*: an encoder is admissible for a required observable Y exactly when it never merges two admissible histories with different Y (Theorem 20), and spatial aliasing at caustics, quantization aliasing at finite resolution, and the closed-orbit obstruction are that one condition violated at three levels. Read this way, preserving J independent branch identities costs exactly a factor $J^2 = 4^{\log_2 J}$—it is the per-bit price of identity in the address, not a separate phenomenon—and reserving history bits is free whenever calibration, rather than address width, is the binding resource. Synthetic experiments demonstrate that a spatial-only representation can manufacture apparent timing fluctuations by merging distinct ray branches at a caustic, and that finite-resolution phase-space representation converts between-branch variance into within-cell variance by an exactly quantifiable identity. The generator, split rule, partition code, and chart table are released so that these figures are reproducible. We state the limits of what can be established in four days:

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machine-checked verification of the complete encoder beyond the released specification, broadband propagation accuracy, geometric approximation quality, and metrological benefit remain explicit validation obligations.

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1 Introduction

Consider an intern tasked with designing a memory structure for measurements that live in three spatial dimensions and two rotational dimensions, occupying a volume whose geometry becomes progressively more elliptical toward the boundary, with contiguous storage, in Ada, for detecting femtosecond-scale fluctuations in an atomic clock—on a four-day deadline. Every noun in this assignment conceals another discipline: memory conceals topology, ellipsoid conceals differential geometry, optics conceals Hamiltonian mechanics, femtosecond conceals precision metrology, and Ada conceals formal verification.

The purpose of this paper is not to discover a new clock. It is to explain why a wrong memory layout can invent one. The surprising central idea is that the correct memory addresses may need to follow the geometry of light propagation, rather than the coordinates occupied by matter. The central claim is not that a five-dimensional array is better than a three-dimensional array. It is that the optimal representation is not the one that stores the fewest coordinates or minimizes a single timing error; it is the least expensive representation that preserves every distinction the instrument may need to make.

1.1 The design problem, formally

The paper's main technical contribution can be stated as a single constrained optimization problem before any of its components are derived. Let Q be an optical state drawn from the instrument's operating distribution μ on the fixed-frequency manifold $\mathcal{M}_{\omega_0}$; let $T_g(Q)$ be its group-delay correction; and let $B_{\mathcal{P}}(Q)$ denote the cell of a partition $\mathcal{P} = \{C_1, \ldots, C_K\}$ containing Q. The design problem is

$$
\begin{aligned}
\min_{\mathcal{P},\{m_k\}} \quad & \mathbb{E}[\mathrm{Var}(T_g(Q) \mid B_{\mathcal{P}}(Q))] + \sum_{k=1}^{K} \frac{p_k v_k}{m_k} \\
\text{subject to} \quad & |\mathcal{P}| \le K_{\max}, \\
& \textstyle\sum_{k=1}^{K} m_k \le M, \\
& \text{each cell has one canonical chart owner and one address,} \\
& \text{addresses are contiguous and collision-free (Theorem 6),} \\
& \text{required ray-branch distinctions are preserved (Theorem 11),} \\
& \text{the chart table is frozen per acquisition batch,}
\end{aligned}
\tag{1}
$$

where $p_k = \mathbb{P}(Q \in C_k)$, $v_k = \mathrm{Var}(T_g \mid C_k)$, $K_{\max}$ is the address budget, and M is the calibration budget. The objective's first term is the geometric distortion resolved by the partition (exactly decomposed by the law of total variance, §11); the second is the statistical cost of estimating each cell's correction from m_k calibration samples. The constraints encode the architectural requirements that are not tradeable: representation correctness (contiguity, uniqueness, frozen versioning) and preservation of the optical distinctions the measurement needs. Everything that follows—the manifold geometry, the encoder, the rate-distortion analysis, and the experiments—is in service of this problem. Making it explicit up front also makes the paper's central claim precise: the optimum is not the partition that minimizes timing distortion alone; it is the cheapest partition in the feasible set.

1.2 Evidence taxonomy

Throughout, we keep three kinds of evidence distinct, because confusing them is how prototypes become folklore.

(A) **Mathematical identities.** Proved: Snell's law from Fermat's principle, the law of total variance, the range bound of the encoder.

(B) **Synthetic numerical results.** Demonstrated under stated models with stated sample sizes. These are not physical atomic-clock measurements; they establish failure modes in controlled settings.

(C) **Formally verified implementation claims.** These require a proof transcript, e.g. a GNATprove transcript. *A transcript is presented in Appendix A for the released encoder, its chart table, and its contract structure only.* Exhaustive enumeration of a concrete chart table is evidence for that table only; it is not a general proof, and the general statement remains Theorem 6.

When we say "proved" we mean (A); when we say "the simulation shows" we mean (B); when we say "we have not proved," category (C) is missing. This distinction is the paper's immune system.

1.3 Related work

The architectural roots of this problem are cross-disciplinary. *Optics and mechanics:* Fermat's principle and the eikonal equation are treated authoritatively by Born and Wolf [1]; Hamiltonian characteristics, phase-space flows, symplectic geometry, and Liouville's theorem by Arnold [2]; caustics, singular ray projections, and Airy/Pearcey catastrophe optics by Berry and Upstill [3–5]; and the contact-geometric structure of the fixed-frequency hypersurface by Geiges [6]. *Metrology:* the statistics of atomic frequency standards under noise perturbations whose character depends on sampling time are due to Allan [7]; this grounds the measurement model of §9.5 and the spectral diagnosis proposed in §14. *Information theory:* Shannon's coding theorems with a fidelity criterion provide the rate-distortion foundation [8]; Lloyd [9] and Max [10] establish optimal scalar quantization; Gray and Neuhoff [11] connect quantization, distortion, resolution, and coding rate; and Misra, Goyal, and Varshney [12] study quantizers designed to minimize the error of a *computed function*—the closest established literature to our task-aware timing objective, with branch constraints and calibration costs specifying the optical specialization. Stone [13] supplies the nonparametric estimation rates underlying the $M^{1/3}$ scaling law of Theorem 12. *Optimization and verification:* the inverse-Hessian sensitivity of Theorem 13 is an application of perturbation analysis of parameterized optimization [14]; and SPARK's contract-based verification methodology is codified by McCormick and Chapin [15]. The paper also builds on the author's prior computational frameworks—physics-informed combinatorial solvers using spectral, topological, and arithmetic methods [16–21], sharing the theme of phase-transition detection and branch-aware optimization—and earlier applied work on content-based retrieval [22]; these are self-citations of record used for context only, not as evidence of type (A), (B), or (C).

1.4 Contributions

(i) A geometric model of progressively ellipsoidal domains with explicit nesting conditions (§3), and the observation that compactness is a storage property while locality is a geometric property, which are not interchangeable (§2).

(ii) Derivation of the optical state space from Fermat's principle, the eikonal equation, and Hamiltonian characteristics; identification of caustics as the failure of *spatial* invertibility, not of ray dynamics (§4–§5).

(iii) A unifying criterion for measurement representations (Appendix E): an encoder is observable sufficient for Y iff equal encodings imply equal Y, under which the caustic failure, finite-resolution aliasing and the closed-orbit obstruction are one condition at three levels, and under which the branch-penalty J^2 of Theorem 11 is the per-bit cost of identity, $J^2 = 4^{\log_2 J}$.

(iv) A chart-aware contiguous encoder with explicit range and injectivity obligations, including the distinction between the two (§6, §7), released as a verification-ready specification with a GNATprove transcript (Appendix A).

(v) Phase-versus-group-delay separation: the state manifold is generated by the phase dispersion relation, while group delay is a different observable computed along its rays (§9).

(vi) A task-aware rate-distortion formulation of cell allocation under memory and calibration budgets (§11), and five synthetic experiments demonstrating concrete failure modes, including a false clock manufactured by spatial projection (§12).

(vii) An explicit statement of what remains unproved: the SPARK transcript, the geometric preprocessing quality, and the metrological benefit (§15).

2 Why an ordinary array is the wrong starting point

A conventional multidimensional array is a convenient interpretation of linear memory. Let

$$(i, j, k), \qquad 0 \le i < N_x, \quad 0 \le j < N_y, \quad 0 \le k < N_z. \tag{2}$$

A row-major representation uses

$$A(i, j, k) = (iN_y + j)N_z + k. \tag{3}$$

This is simple, collision-free, and invertible. Adding two angular coordinates discretized into N_θ and N_ϕ bins yields

$$A(i, j, k, p, q) = ((((iN_y + j)N_z + k)N_\theta + p)N_\phi + q. \tag{4}$$

Five dimensions have become one integer. Problem solved? Not remotely.

This formula assumes that the logical domain is a Cartesian product $X \times Y \times Z \times \Theta \times \Phi$. But an ellipsoidal volume is not a rectangular box, and a progressively deforming ellipsoidal volume is not naturally described by uniformly sampled Cartesian slices. Allocating the bounding box wastes memory outside the valid domain; enumerating valid lattice points with consecutive ranks saves memory but does not guarantee that neighboring optical states are stored near one another.

A *compact address map* answers: can I store every admissible state without wasting addresses? An optical memory architecture must answer a harder question: can I store states so that the geometry relevant to the measurement survives discretization? Compactness is a storage property; locality is a geometric property. They are not interchangeable.

Consider two measurements that are close in Euclidean space but arrive along different ray branches with very different propagation delays. A compact spatial rank puts them together; an optical memory that cares about timing should not necessarily. Conversely, two measurements far apart in space but sharing the same group delay may be natural comparison partners for timing analysis; a spatial layout separates them. The intern's first mistake would be to make the array small rather than faithful.

3 A progressively elliptical domain

Introduce a normalized radial parameter $0 \le \rho \le 1$ and a unit direction $\mathbf{u} \in S^2$. Define

$$\mathbf{x}(\rho, \mathbf{u}) = \rho \begin{pmatrix} a(\rho) & 0 & 0 \\ 0 & b(\rho) & 0 \\ 0 & 0 & c(\rho) \end{pmatrix} \mathbf{u}, \qquad a(\rho) = 1 + \alpha\rho^2, \quad b(\rho) = 1, \quad c(\rho) = 1 - \beta\rho^2, \tag{5}$$

with restrictions

$$\boxed{\alpha > 0, \qquad 0 < \beta < \tfrac{1}{3}.} \tag{6}$$

These are not cosmetic. The semiaxes of the shell at radial parameter ρ are

$$s_x(\rho) = \rho + \alpha\rho^3, \qquad s_y(\rho) = \rho, \qquad s_z(\rho) = \rho - \beta\rho^3. \tag{7}$$

For the ellipsoids to remain nested, their semiaxes must increase with ρ.

Proposition 1 (Nesting)**.** *For $\alpha > 0$ and $0 < \beta < 1/3$, the family $\{\mathbf{E}_\rho\}_{0 \le \rho \le 1}$ is strictly nested: each shell encloses the preceding one, and the construction extends continuously to the center, where the shell at $\rho = 0$ degenerates to a point.*

Proof. Differentiate: $\frac{ds_x}{d\rho} = 1 + 3\alpha\rho^2 > 0$, $\frac{ds_y}{d\rho} = 1 > 0$, and $\frac{ds_z}{d\rho} = 1 - 3\beta\rho^2 > 0$, the final inequality following from $\beta < 1/3$. Strictly increasing semiaxes imply each shell is interior to the next. □

Near the center the three axes are approximately equal; near the boundary their ratios differ. We have a family of increasingly elliptical shells.

However, this is only a geometric model. It does not establish that light propagates along those shells, or that the shells are surfaces of equal optical travel time. To determine that, we need optics—and a seventeenth-century mathematician enters the data-structure design meeting.

4 Fermat's principle as an address generator

Fermat's principle states that physical rays follow paths for which optical travel time is stationary under suitable variations [1, 23]. For a refractive-index field $n(\mathbf{x})$, the travel-time functional is

$$\mathcal{T}[\gamma] = \frac{1}{c} \int_\gamma n(\mathbf{x})\, d\ell, \tag{8}$$

where ℓ is physical arc length. A physical ray satisfies $\delta\mathcal{T} = 0$; "stationary" is more precise than "minimum," since a ray path need not be a strict global minimum.

Why should a memory engineer care? Because the device is not primarily interested in where a measurement sits in Euclidean space; it is interested in the phase and arrival time of optical information. Two spatially nearby measurements can have different propagation delays; two spatially distant measurements can share the same optical phase or travel-time surface.

Introduce a phase-travel-time field $T_p(\mathbf{x})$. For isotropic geometric optics at a fixed frequency, away from singularities, it satisfies the eikonal equation:

$$\boxed{|\nabla T_p(\mathbf{x})|^2 = \frac{n_p^2(\mathbf{x}, \omega_0)}{c^2}.} \tag{9}$$

The level sets $\Sigma_\tau = \{\mathbf{x} : T_p(\mathbf{x}) = \tau\}$ are equal-phase wavefronts at the chosen frequency. The memory problem now changes character: instead of asking which Cartesian cells belong inside an

ellipsoid, we ask which optical states belong to each propagation-related layer. If the actual optical medium produces increasingly elliptical wavefronts, the resulting memory organization can reflect that deformation.

Remark 2. *The conditional matters: progressive ellipticity must come from the optical model or measured calibration data. Fermat's principle does not guarantee it. Using optical propagation geometry as an organizing principle, rather than treating the ellipsoidal boundary as a Boolean mask, is our first architectural improvement—and it creates a new problem: what happens when two rays arrive at the same place?*

4.1 The Cauchy problem, or why rays refuse to behave like array indices

Specify an initial wavefront with its initial phase and propagation directions; this is a Cauchy problem for subsequent propagation. The eikonal equation belongs to the Hamilton–Jacobi family [2, 24]; its characteristic curves derive from a Hamiltonian. Fix the optical frequency and suppress it from the notation. Define

$$H(\mathbf{x}, \mathbf{p}) = \frac{1}{2}\left(|\mathbf{p}|^2 - \frac{n_p^2(\mathbf{x})}{c^2}\right). \tag{10}$$

Optical rays lie on $H = 0$ and evolve by Hamilton's equations:

$$\frac{d\mathbf{x}}{ds} = \frac{\partial H}{\partial \mathbf{p}} = \mathbf{p}, \tag{11}$$

$$\frac{d\mathbf{p}}{ds} = -\frac{\partial H}{\partial \mathbf{x}} = \frac{n_p(\mathbf{x})}{c^2}\nabla n_p(\mathbf{x}), \tag{12}$$

where s is a characteristic parameter, not physical time. On $H = 0$, $|\mathbf{p}| = n_p/c$, hence $|d\mathbf{x}/ds| = n_p/c$ and $ds = (c/n_p)\, d\ell$, giving

$$\boxed{s - s_0 = c\int_{\ell_0}^{\ell} \frac{d\ell'}{n_p(\ell')}.} \tag{13}$$

This parameter is neither physical time nor optical travel time $T_p = \int \frac{n_p}{c}\, d\ell$; its normalization changes traversal speed but not the geometric ray trajectory.

Given an initial wavefront parameterized by (u, v), the ray family $\mathbf{x}(s, u, v)$ locally provides three coordinates for three-dimensional space. That looks promising. Unfortunately, nature has invented caustics.

4.2 The caustic problem

When sunlight passes through a glass of water, bright curves appear on the table because neighboring rays concentrate. Mathematically, the map $(s, u, v) \mapsto \mathbf{x}(s, u, v)$ may cease to be locally invertible. Define the Jacobian

$$J = \det\left[\frac{\partial \mathbf{x}}{\partial s}, \frac{\partial \mathbf{x}}{\partial u}, \frac{\partial \mathbf{x}}{\partial v}\right]. \tag{14}$$

Proposition 3 (Caustic signature)**.** *The spatial ray-coordinate map loses local invertibility exactly where $J = 0$.*

Proof. Immediate from the inverse function theorem: local invertibility of the parametrization requires a nonsingular Jacobian matrix. □

The key observation: the rays themselves need not cease to exist. What fails is our attempt to identify a ray using spatial position alone. Suppose two rays reach the same spatial point, $\mathbf{x}_1 = \mathbf{x}_2$; they can still have different momenta, $\mathbf{p}_1 \neq \mathbf{p}_2$. A spatial-only address merges them. An optical state-space address can keep them separate. This is why the two rotational degrees of freedom in the original specification are potentially far more important than they first appear: they can encode the propagation direction that distinguishes ray branches.

5 The five-dimensional escape hatch

In ordinary Hamiltonian mechanics, a particle in three-dimensional space has a six-dimensional phase space (x, y, z, p_x, p_y, p_z). In our isotropic fixed-frequency optical model, the Hamiltonian constraint determines the magnitude of optical momentum,

$$|\mathbf{p}| = \frac{n_p(\mathbf{x}, \omega_0)}{c}, \tag{15}$$

leaving only its direction free. A direction in three dimensions requires two angular coordinates:

$$\mathbf{p} = \frac{n_p(\mathbf{x}, \omega_0)}{c} \begin{pmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{pmatrix}. \tag{16}$$

We arrive at the boxed state tuple

$$\boxed{(x, y, z, \theta, \phi)} \tag{17}$$

—three spatial dimensions, two rotational dimensions—exactly the five degrees of freedom from the assignment, under the interpretation that the rotations are propagation directions. This is the most natural reading for fixed-frequency propagation in an isotropic medium, but not the only physical interpretation: a polarization-frame orientation or an aperture sampling scheme could introduce different angular variables. Our construction specifically uses propagation direction because it distinguishes intersecting ray branches.

The natural mathematical object is the fixed-frequency constraint manifold

$$\mathcal{M}_{\omega_0} = \left\{ (\mathbf{x}, \mathbf{p}) : |\mathbf{p}| = \frac{n_p(\mathbf{x}, \omega_0)}{c} \right\}. \tag{18}$$

Proposition 4. *For a smooth positive refractive index n_p, $\mathcal{M}_{\omega_0}$ is a five-dimensional manifold, and the projection $\pi : \mathcal{M}_{\omega_0} \to \mathbb{R}^3$ may fail to distinguish distinct states: two rays can occupy the same spatial point but remain distinct points of $\mathcal{M}_{\omega_0}$ because their directions differ.*

Proof. $F(\mathbf{x}, \mathbf{p}) = |\mathbf{p}|^2 - n_p(\mathbf{x}, \omega_0)^2/c^2$ satisfies $\partial_{\mathbf{p}} F = 2\mathbf{p} \neq 0$ on $F = 0$ (since $n_p > 0$ forces $|\mathbf{p}| > 0$), so the regular level set theorem applies in the six-dimensional ambient space, yielding dimension 5. The failure of π to be injective is witnessed by any pair $(\mathbf{x}, \mathbf{p}_1) \neq (\mathbf{x}, \mathbf{p}_2)$ with $|\mathbf{p}_i| = n_p(\mathbf{x}, \omega_0)/c$ and $\mathbf{p}_1 \neq \mathbf{p}_2$, which exists whenever $\mathbf{x}$ admits two distinct propagation directions. □

Remark 5 (Why multiple charts)**.** *A five-dimensional coordinate chart is not globally nonsingular: angular coordinates have poles and seams, and complicated geometry may require multiple charts. Nor is every ray branch uniquely identified by quantized position and direction: distinct samples can occupy the same discrete cell, requiring additional sample sequence or branch identity. Nevertheless the correct basic representation has emerged: store optical states on the constrained phase-space manifold, rather than storing spatial cells and attaching orientation as an afterthought.*

6 Constructing the contiguous address space

We want a single memory arena $M_0, M_1, \ldots, M_{N-1}$—no giant five-dimensional rectangular allocation. Instead, construct a finite partition of the admissible optical state manifold into cells $C_0, C_1, \ldots, C_{N-1}$, each representing a small region of position–direction space. The encoder is a map

$$E : \mathcal{M}_{\text{discrete}} \to \{0, \ldots, N-1\}. \tag{19}$$

We require *uniqueness* ($E(q_1) = E(q_2) \implies q_1 = q_2$ for canonical discrete cell identities), *coverage* (every admitted cell has an address), *stability* (the map remains well-defined near caustics and deformed boundaries), and, where practical, *geometric locality* (optically nearby states occupy nearby addresses). The last requirement is an optimization objective, not a consequence of compact indexing.

6.1 Multiple optical charts

A single global chart is fragile. Divide the manifold into local patches and assign each physical cell a canonical chart owner. For each chart k, store its base offset B_k, cell count N_k, calibration version, and mapping from canonical cell identities to local ranks. Define

$$\boxed{E(k, r) = B_k + r, \qquad 0 \le r < N_k,} \tag{20}$$

with consecutive intervals $B_0 = 0$, $B_{k+1} = B_k + N_k$. This produces contiguous storage without gaps. Charts may overlap geometrically, but their canonical ownership must not, or the same logical cell could acquire multiple addresses.

There is also a crucial distinction between unique cells and unique continuous states: finite digital storage identifies quantized cells, not every real-valued point inside them, and the quantization error must be bounded relative to the measurement requirements.

6.2 The address budget does not survive refinement unchanged

A caustic warns that the spatial projection becomes singular; it does not automatically mean every nearby cell must be refined. Refinement is justified where an error estimator, ray-branch separation criterion, or measurement sensitivity requires it. And refinement is not free: with a fixed address budget, allocating more cells near a difficult region reduces the budget available elsewhere. Two failure modes follow. First, smooth regions become under-resolved, potentially corrupting timing analysis where the system was expected to behave quietly. Second, allocation becomes calibration-dependent: two runs with different optical configurations may generate different chart tables, changing what a given address means—a reproducibility hazard disguised as a memory optimization.

The four-day compromise: fix the global cell budget, allocate it by a documented rule, freeze the resulting chart table during acquisition, and record its version with every measurement batch. Changing the allocation rule is a change to the instrument, not merely a software optimization.

7 Why Ada suddenly makes sense

At this point a conventional programmer might say, "Fine, let's put everything in a hash table." The senior infrastructure engineer asks more unpleasant questions: can you prove the encoder never

produces an invalid address? Can you prove two distinct canonical cells never alias? What happens when an interrupt arrives during a geometry-table update?

Now Ada becomes interesting. Ada offers strong typing, constrained ranges, and contracts [25]; SPARK, its formally analyzable subset, can establish selected program properties [15, 26]. For precision instrumentation this matters because a silent indexing error can produce a plausible but physically meaningless measurement. We do not merely want the program to avoid crashing; we want it to avoid silently turning one optical state into another.

7.1 The encoder's proof obligations

Assume an offline geometry preprocessor has constructed canonical charts and assigned local ranks. The essential range obligation is easy to understand. We need $B_k + r < N$ for every valid rank r. Given $B_k + N_k \leq N$, we obtain $B_k + r < B_k + N_k \leq N$, so the address is in bounds.

Proof Obligation 1 (Range bound)**.** *For all valid chart identifiers k and ranks $0 \leq r < N_k$: $B_k + r < N$.*

But range safety is not injectivity.

Theorem 6 (Injectivity)**.** *If the chart intervals $[B_k, B_k + N_k)$ are pairwise disjoint, then $E(k, r) = B_k + r$ is injective on $\{(k, r) : 0 \leq r < N_k\}$; if the intervals additionally cover the complete address range consecutively, E is a bijection onto that range.*

Proof. Suppose $E(i, r) = E(j, s)$. If $i \neq j$, disjoint intervals make equality impossible, so $i = j$; then $B_i + r = B_i + s$ implies $r = s$. Surjectivity follows from the consecutive-covering hypothesis by induction over the interval sequence. □

Remark 7. *These are two related but distinct proof obligations. The range bound concerns a single call to* `Encode`*; injectivity compares arbitrary pairs of valid inputs and is a* relational *property requiring its own proof, whether by a mathematical argument or a SPARK ghost lemma. The released specification (Appendix A) supplies the invariant structure, and its GNATprove transcript discharges all 47 obligations with no unproved checks: range safety, initialization, and the functional contracts of* `Encode`*,* `Decode`*, and the ghost injectivity lemma. This machine-checked result applies to the released chart table and contract structure; it is not claimed for arbitrary chart tables or for the geometry preprocessor, and the general statement remains the mathematical Theorem 6.*

There is also a deeper missing proof: nothing in this encoder establishes that the geometry preprocessor constructed the correct optical partition. The solver establishes what the cells mean; the verification-ready encoder establishes where they live; the acquisition pipeline establishes which measurements belong to them. Keeping these responsibilities separate is what makes the system auditable. Formal verification is not a property of the author's intentions; it is a claim that must survive actual proof obligations—which is why the transcript is released alongside the specification.

8 The IIT-JEE-style mathematical hammer

We temporarily forget the terminology and attack the central ideas with calculus, vectors, and algebra.

8.1 Deriving Snell's law from Fermat's principle

Light travels from point A in a medium of refractive index n_1 to point B in a medium of index n_2, through a horizontal interface. Let h_1, h_2 be the perpendicular distances, L the horizontal separation, and x the crossing coordinate. Travel time:

$$T(x) = \frac{n_1}{c}\sqrt{x^2 + h_1^2} + \frac{n_2}{c}\sqrt{(L-x)^2 + h_2^2}. \tag{21}$$

Differentiating and setting to zero:

$$\frac{n_1 x}{\sqrt{x^2 + h_1^2}} = \frac{n_2(L-x)}{\sqrt{(L-x)^2 + h_2^2}}, \tag{22}$$

and recognizing $\sin\theta_1 = \frac{x}{\sqrt{x^2+h_1^2}}$, $\sin\theta_2 = \frac{L-x}{\sqrt{(L-x)^2+h_2^2}}$:

$$\boxed{n_1 \sin\theta_1 = n_2 \sin\theta_2.} \tag{23}$$

Snell's law appears directly from stationary travel time. The lesson: physically meaningful trajectories arise from the optical variational principle, and a programmer's coordinate grid has no reason to respect those trajectories automatically.

8.2 Why a caustic destroys spatial invertibility

Consider the two-dimensional toy family of rays

$$x(t,u) = u(1-t), \qquad y(t,u) = t, \tag{24}$$

where u labels the initial ray and t parameterizes its evolution. At $t = 1$, $x(1,u) = 0$ for every u: all rays meet at $(0,1)$. The Jacobian is

$$J = \det\begin{pmatrix} \partial_t x & \partial_u x \\ \partial_t y & \partial_u y \end{pmatrix} = \det\begin{pmatrix} -u & 1-t \\ 1 & 0 \end{pmatrix} = t - 1, \qquad \boxed{J = 0 \text{ at } t = 1.} \tag{25}$$

The inverse mapping fails precisely at the focus: away from $t = 1$, $u = \frac{x}{1-y}$, whose denominator vanishes at $y = 1$. This is a mathematical failure of the coordinates, not a floating-point accident. Now examine the ray tangent $\frac{\partial(x,y)}{\partial t} = (-u, 1)$: at the focus, rays with different u have different directions. Their spatial positions coincide, but their lifted position–direction states remain distinguishable—which is why storing orientation alongside position preserves information that a spatial address loses.

8.3 Quantifying the danger of a progressively elliptical boundary

Return to the parametrization of §3: $x = \rho(1 + \alpha\rho^2)u_x$, $y = \rho u_y$, $z = \rho(1 - \beta\rho^2)u_z$. Along the positive x-axis, $x(\rho) = \rho + \alpha\rho^3$ and $\frac{dx}{d\rho} = 1 + 3\alpha\rho^2$, approaching 1 at the center and $\left.\frac{dx}{d\rho}\right|_{\rho=1} = 1 + 3\alpha$ at the boundary; for $\alpha = 0.2$ this is 1.6. A fixed radial increment produces approximately 60% more physical displacement near the outer boundary along this axis than near the center. Along the z-axis, $z(\rho) = \rho - \beta\rho^3$ and $\frac{dz}{d\rho} = 1 - 3\beta\rho^2$; for $\beta = 0.2$ the boundary value is 0.4. Uniform increments in ρ are not uniform physical distances, nor necessarily uniform optical travel-time increments.

The enclosed volume is

$$V(\rho) = \frac{4\pi}{3}\rho^3 a(\rho) b(\rho) c(\rho), \tag{26}$$

and the nesting conditions of Proposition 1 ensure these are actual nested ellipsoids with increasing enclosed volumes. Differentiating:

$$\frac{dV}{d\rho} = 4\pi\rho^2 abc + \frac{4\pi}{3}\rho^3(a'bc + ab'c + abc'), \tag{27}$$

whose first term resembles ordinary radial growth and whose second term comes from the changing ellipsoidal shape. A shell-indexing scheme that ignores this contribution misrepresents the volume distribution—before accounting for refractive index, ray focusing, or directional sampling. A compact spatial-rank lookup can give every cell a unique number; it cannot, by itself, ensure those numbers respect the physics.

9 The femtosecond connection

One femtosecond is 10^{-15} s; light travels $c\Delta t = (3 \times 10^8)(10^{-15}) = 3 \times 10^{-7}$ m = $\boxed{300\,\text{nm}}$ during that interval. A propagation-delay error of this scale can contaminate a measurement intended to resolve femtosecond timing structure.

9.1 Phase velocity, group velocity, and why one refractive index is not enough

The eikonal equation of §4 uses the phase refractive index n_p, whose level sets describe equal phase. For a continuous-wave interferometric measurement, phase is a natural observable. For a pulse-arrival measurement, however, group delay becomes important. In a time-independent isotropic medium the dispersion relation is

$$|\mathbf{k}| = \frac{n_p(\mathbf{x},\omega)\omega}{c}, \qquad v_p = \frac{\omega}{|\mathbf{k}|} = \frac{c}{n_p}, \tag{28}$$

and differentiating,

$$\boxed{v_g = \frac{\partial\omega}{\partial|\mathbf{k}|} = \frac{c}{n_p + \omega\partial_\omega n_p} = \frac{c}{n_g},} \qquad n_g = n_p + \omega\partial_\omega n_p. \tag{29}$$

It would be tempting to replace n_p with n_g inside the eikonal Hamiltonian. That is generally wrong: the phase dispersion relation determines the rays; the group index is derived from that dispersion relation, not a replacement for it.

9.2 The extended Hamiltonian

Introduce

$$\boxed{\mathcal{H}(\mathbf{x},\mathbf{k},\omega) = c^2|\mathbf{k}|^2 - n_p^2(\mathbf{x},\omega)\omega^2 = 0,} \tag{30}$$

a seven-dimensional description with a codimension-one dispersion constraint. For a canonical Hamiltonian treatment of propagation and timing, include the time coordinate t conjugate to ω, giving an eight-dimensional extended phase space before imposing the constraint. Because the medium is time-independent, frequency is conserved along rays, $d\omega/d\tau = 0$; at fixed ω_0 the dispersion constraint reduces to the five-dimensional manifold of §5. Group delay is computed along the phase-generated ray trajectories: for a narrowband pulse propagating along a specified ray branch γ,

$$\boxed{T_g(A,B;\omega_0) = \int_\gamma \frac{n_g(\mathbf{x},\omega_0)}{c}\,d\ell,} \tag{31}$$

assuming an appropriate isotropic stationary-medium setting and a well-defined ray branch. The state manifold remains phase-defined; what changes is the functional used to organize measurements. A group-delay-sensitive memory layout may benefit from grouping states according to T_g rather than T_p. State space and organizing coordinate are separate architectural decisions.

9.3 Caustics do not move merely because the observable changes

Changing the indexing functional from phase delay to group delay does not change the physical ray family at a fixed frequency, and therefore does not independently move the physical caustics, which are determined by the spatial projection of the phase-generated rays. However, a group-delay coordinate may introduce its own representation difficulties: it may fail to distinguish certain states or become multivalued when projected onto physical space. Those are coordinate or branch-selection problems, not new physical caustics. A caustic itself is not necessarily a breakdown of ray dynamics—rays can remain smooth while their spatial projection becomes singular. What fails in elementary geometrical optics is often the naive amplitude approximation, which can diverge at an ideal caustic; uniform asymptotic methods including Airy-type approximations for folds [5] and Pearcey-type approximations for cusps [3, 4] describe caustic neighborhoods without a complete Maxwell simulation. When wavelength-scale material structure or strong scattering invalidates the approximation, more complete wave modeling may be necessary.

9.4 The bandwidth problem

The time-bandwidth product is constrained by the pulse shape: $\Delta\omega\,\Delta t \gtrsim O(1)$, with the numerical constant depending on how duration and spectral width are defined. At optical frequencies, a transform-limited pulse lasting a few femtoseconds has substantial relative bandwidth; a chirped pulse of the same temporal duration can have a different spectral width, and chirp alone does not establish a narrower bandwidth. For a broadband pulse, evaluating n_g only at ω_0 may be inadequate: higher-order dispersion, spectral phase, and pulse reshaping can affect the measured arrival time. The four-day prototype therefore assumes a narrowband regime, uses T_g as a leading-order organizing functional, and explicitly records that this approximation may not hold for the intended instrument. If the device sweeps frequency, chart tables may need versioning per band, or frequency may need to become an additional state coordinate.

The compact lesson: the phase dispersion relation generates the rays; group delay describes a timing observable along those rays; organizing memory by group delay changes the indexing geometry, not the underlying ray dynamics. Confusing these roles would mean building the correct data structure around the wrong physical quantity.

9.5 The measurement model

A simplified timing measurement can be written [7]

$$t_{\mathrm{obs}} = t_{\mathrm{clock}} + T_g + t_{\mathrm{electronics}} + \epsilon. \tag{32}$$

After calibration,

$$r = t_{\mathrm{obs}} - \widehat{T}_g - \widehat{t}_{\mathrm{electronics}}, \tag{33}$$

and the objective is to investigate whether r contains structure beyond expected noise and known clock behavior. Our memory architecture can help preserve the information required for propagation corrections and comparisons; it cannot manufacture measurement precision. If the detector,

digitizer, reference oscillator, or acquisition electronics cannot resolve the relevant fluctuations, a more sophisticated address mapping cannot recover information that was never captured.

There is also a subtle question of locality. Two samples may be spatially distant yet share similar calibrated group delays, making them natural comparison partners for some analyses. A possible analysis-oriented distance is

$$d^2(q_1, q_2) = (T_1 - T_2)^2 + w_\Omega d^2_{S^2}(\Omega_1, \Omega_2) + \frac{\|\mathbf{x}_1 - \mathbf{x}_2\|^2}{v_g^2}, \tag{34}$$

where w_Ω has units of time squared. This is a design metric, not a physical law: the weights encode which differences matter for a particular analysis, must be calibrated against the measurement objective, and should not be mistaken for universal constants of optical geometry.

10 What can actually be built in four days?

A four-day prototype should establish the architecture's critical invariants and expose unresolved physics rather than concealing it. The geometry computation need not initially run on the SoC: it can be performed offline and exported as immutable calibration tables, separating an expensive numerical solver from a bounded runtime address lookup. We distinguish three levels of success.

Tier 1: Representation correctness. Every canonical cell has one valid address, and decoding reconstructs its identity.

Tier 2: Geometric correctness. The cells cover the intended optical state space with quantified approximation errors, including near difficult boundaries and caustics.

Tier 3: Measurement usefulness. The representation improves or preserves timing-analysis accuracy relative to a suitable baseline.

Day four can demonstrate Tier 1 on a completed test implementation and begin evaluating Tier 2 on synthetic data; Tier 3 is explicitly outside the four-day scope. Passing Tier 1 does not imply Tier 2, and passing Tier 2 does not imply Tier 3. The chart-table version must be frozen during an acquisition batch: updating the optical partition mid-measurement could change what an address means, so a calibration version is part of the sample's interpretation even if recorded only once per batch. This is the sort of mundane detail that distinguishes an engineering prototype from a mathematical sketch.

11 From optical geometry to rate-distortion theory

The preceding sections establish that a spatial-only representation can introduce false timing variation, while a finite-resolution phase-space representation preserves much of the missing information—but a new problem emerges: how should a limited number of memory cells be distributed across optical phase space? Uniform spatial sampling is not necessarily appropriate; nor uniform angular sampling; nor even refining wherever the ray-coordinate Jacobian becomes small. The correct allocation depends on what the instrument is trying to measure.

11.1 The loss function

Let Q be an optical state drawn from the instrument's operating distribution μ, and let $T_g(Q)$ denote its group-delay correction. A partition $\mathcal{P} = \{C_1, \ldots, C_K\}$ assigns each optical state to a discrete cell $B_{\mathcal{P}}(Q)$. Within each cell, the optimal constant group-delay estimate under squared error is $\widehat{T}_g(C_k) = \mathbb{E}[T_g(Q) \mid Q \in C_k]$. The resulting distortion is

$$\boxed{\mathcal{L}(\mathcal{P}) = \mathbb{E}[\mathrm{Var}(T_g(Q) \mid B_{\mathcal{P}}(Q))]\,,} \tag{35}$$

so for a fixed SoC cell budget $K_{\max}$ the design problem becomes

$$\boxed{\begin{aligned} \mathcal{P}^\star &= \arg\min_{\mathcal{P}} \mathcal{L}(\mathcal{P}), \\ \text{subject to} \quad & |\mathcal{P}| \le K_{\max}. \end{aligned}} \tag{36}$$

Additional hardware constraints may require contiguous chart regions, bounded lookup latency, canonical ownership, minimum spatial resolution, or fixed per-chart allocations. This is a *task-aware quantization* problem: unlike conventional geometric meshing, its objective is not merely to approximate the shape of the optical manifold but to minimize unresolved timing error.

11.2 Why the variance decomposition matters

The law of total variance gives

$$\mathrm{Var}(T) = \mathrm{Var}(\mathbb{E}[T \mid B]) + \mathbb{E}[\mathrm{Var}(T \mid B)]. \tag{37}$$

The first term measures geometric timing variation resolved by the partition; the second measures timing variation remaining inside cells. Therefore

$$\boxed{\text{Resolved fraction} = 1 - \frac{\mathcal{L}(\mathcal{P})}{\mathrm{Var}(T)}.} \tag{38}$$

In one simulation, the empirical decomposition was $\mathrm{Var}(T) = \mathrm{Var}(\mathbb{E}[T \mid B]) + \mathcal{L}(\mathcal{P})$, and the partition resolved approximately 92% of the simulated geometric-delay variance. This percentage uses the empirical total variance from that experiment and should not be confused with the fraction obtained from earlier finite samples with different residual estimates. The distinction matters because the loss is an expectation over the operating distribution, while experiments only estimate that expectation from finite samples.

11.3 Calibration is part of the budget

The optimal cell correction is a conditional expectation, which in practice must be estimated from calibration data. Suppose cell k contains m_k independent calibration observations with conditional variance v_k, and its correction is estimated by the sample mean. Then $\mathrm{Var}(\widehat{T}_k \mid C_k) = v_k/m_k$, and under independent calibration and evaluation data the expected timing mean-squared error becomes

$$\boxed{\mathrm{MSE} = \sigma^2 + \sum_{k=1}^{K} p_k \left(v_k + \frac{v_k}{m_k} \right),} \tag{39}$$

where $p_k = \mathbb{P}(Q \in C_k)$, $v_k = \mathrm{Var}(T_g \mid C_k)$, and σ^2 is evaluation detector-noise variance. (If calibration measurements also contain detector noise, that contribution must be included in the

calibration estimator variance.) This creates a second resource constraint: more cells can reduce within-cell distortion but also require more calibration observations to estimate their corrections reliably. With a fixed calibration budget M, the design problem becomes a joint optimization over the partition and its calibration allocation:

$$\boxed{\min_{\mathcal{P},\{m_k\}} \left[\mathcal{L}(\mathcal{P}) + \sum_k p_k \frac{v_k}{m_k} \right]} \qquad \text{subject to } |\mathcal{P}| \le K_{\max},\ \sum_k m_k \le M. \tag{40}$$

The instrument must budget both memory and calibration effort: a cell that is cheap to allocate in silicon may be expensive to calibrate.

Proposition 8 (Optimal calibration allocation)**.** *If the cell probabilities p_k and conditional variances v_k are known, then the calibration allocation minimizing the total estimator-variance contribution under a total sample budget M is, in the continuous relaxation,*

$$\boxed{m_k \propto \sqrt{p_k v_k}.} \tag{41}$$

Proof. Lagrange multipliers: minimize $\sum_k p_k v_k / m_k$ subject to $\sum_k m_k = M$. The stationarity condition $-p_k v_k / m_k^2 + \lambda = 0$ gives $m_k = \sqrt{p_k v_k / \lambda}$; summing and enforcing the constraint fixes λ. □

So even calibration rays should not necessarily be allocated uniformly—a direct consequence of the multiplier calculation.

11.4 The central optimization

Putting it all together, with the constraints explicit:

$$\boxed{\begin{aligned} \min_{\mathcal{P},\{m_k\}} \quad & \mathbb{E}[\mathrm{Var}(T_g \mid B_{\mathcal{P}})] + \sum_k \frac{p_k v_k}{m_k} \\ \text{subject to} \quad & |\mathcal{P}| \le K_{\max}, \\ & \sum_k m_k \le M, \\ & \mathcal{P} \text{ preserves chart ownership}, \\ & \mathcal{P} \text{ preserves required branch identity}, \end{aligned}} \tag{42}$$

with contiguous angular cells as an additional constraint for the dynamic-programming experiment of §12. This formulation makes the central tradeoff explicit: timing distortion, memory capacity, calibration effort, and optical identity cannot be optimized independently. For scalar T_g with squared error, the unconstrained problem is Lloyd–Max quantization [9, 10]; with contiguity, chart ownership, and branch-identity constraints, it becomes constrained weighted quantization. Lloyd–Max provides the foundation, not automatically the global solution: the unconstrained optimum may merge distinct branches, which is why it cannot be deployed even when its timing number looks better.

The classical rate-distortion problem asks how much information must be retained to achieve a desired reconstruction accuracy [8, 27]; here the distortion measure is physically meaningful: unresolved group-delay variation. The hardware imposes a finite set of addresses, the optical model imposes geometric constraints, and calibration imposes a statistical cost. The resulting partition is not merely a data structure; it is part of the measurement system.

12 Experimental results: when the wrong address becomes a false clock

We evaluated five synthetic experiments designed to isolate the consequences of spatial projection, anisotropic geometry, dispersive propagation, chart-aware memory addressing, and task-aware allocation. All are evidence type (B) of §1.2, except where an identity is marked (A).

12.1 Experiment 1: A caustic can manufacture timing structure

We generated 20,000 rays converging to a common spatial location. Each ray retained a distinct propagation direction and a direction-dependent group delay. Let U denote the ray's initial direction parameter, and let the measured arrival time be

$$Y = T_g(U) + \varepsilon, \tag{43}$$

where ε is independent detector noise with $\mathbb{E}[\varepsilon] = 0$, $\mathrm{Var}(\varepsilon) = \sigma^2$. A spatial-only representation merges rays arriving at the same position; if the reconstruction subtracts a single mean delay, its residual is $R_{\text{merged}} = T_g(U) - \mathbb{E}[T_g(U)] + \varepsilon$. By independence,

$$\boxed{\mathrm{Var}(R_{\text{merged}}) = \mathrm{Var}_U(T_g) + \sigma^2.} \tag{44}$$

The first term is not clock noise; it is unresolved geometric delay variation. A phase-space representation retains ray direction, allowing the correct branch-dependent delay to be subtracted:

$$R_{\text{lifted}} = Y - T_g(U) = \varepsilon, \qquad \boxed{\mathrm{Var}(R_{\text{lifted}}) = \sigma^2.} \tag{45}$$

In the simulation, $\mathrm{std}[T_g(U)] = 2.85\,\mathrm{fs}$, $\sigma = 0.30\,\mathrm{fs}$, ideal merged residual std $= 2.87\,\mathrm{fs}$, variance $= 8.21\,\mathrm{fs}^2$. The binned lifted residual with $K = 128$ bins had std $= 0.34\,\mathrm{fs}$, variance $= 0.117\,\mathrm{fs}^2$. Equivalently, under the model, $\sqrt{\mathrm{Var}_U(T_g) + \sigma^2}/\sigma = 9.55$: the merged residual is $9.55\times$ the detector-noise std, $91\times$ the variance. Compared to the practical binned lifted residual, it is $8.34\times$ the std, $69.6\times$ the variance. The difference between those ratios matters: the first compares to the ideal floor $\sigma = 0.30\,\mathrm{fs}$; the second compares to the practical floor $\sqrt{\sigma^2 + \mathrm{Var}(T_g \mid B)} \approx 0.34\,\mathrm{fs}$. The binned lifted residual is not σ; it is σ plus within-bin geometry $\mathrm{Var}(T_g \mid B)$. Thus the binned representation resolves $1 - 0.027/8.12 = 99.7\%$ of the geometric-delay variance in that sample, leaving $\mathrm{Var}(T_g \mid B) = 0.027\,\mathrm{fs}^2$ unresolved, i.e. approximately 0.3% unresolved.

The experiment demonstrates a specific failure mode: spatial merging can turn deterministic, branch-dependent propagation delays into apparent stochastic timing fluctuations. It does not demonstrate a new clock; it demonstrates how a false clock-like signal can arise.

Figure 1 (numerical summary). Histograms of spatially merged and phase-space-corrected timing residuals: the merged distribution is broad and asymmetric (skew 0.42, excess kurtosis -0.85); the corrected distribution is concentrated near the detector-noise scale but remains binning-limited (skew 0.05, excess kurtosis 0.51); the Shapiro–Wilk test [28] rejects exact normality on $n = 20{,}000$ at $p < 10^{-6}$. Call it noise-and-binning-limited, not an honest Gaussian.

12.2 Experiment 2: Uniform shell indices are not uniform physical sampling

We evaluated the progressively ellipsoidal shell family of §3 with $\alpha = 0.2$, $\beta = 0.2$. At the outer boundary, $dx/d\rho = 1.6$ and $dz/d\rho = 0.4$: the selected parameters produce a 4:1 ratio between the

physical displacements generated by equal radial increments along the two axes. (At $\rho = 0.5$ the ratio is 1.15:0.85 $\approx$ 1.35, recovering ≈ 1 in the central nearly spherical region.) Therefore uniformly spaced ρ-bins do not correspond to uniform physical sampling.

This is not an address-collision problem: every cell may still have a unique integer address. It is a geometric-resolution problem: identical index increments represent substantially different physical displacements. A chart allocator must therefore account for geometric distortion, using Jacobian information or an appropriate error-based refinement criterion. Volume check: $V(1) = \frac{4\pi}{3}(1.2)(1)(0.8) \approx 4.82$ versus sphere $\frac{4\pi}{3} \approx 4.19$; the mean-radius cube estimate $((1.2+1+0.8)/3)^3 \approx 1$ yields the corresponding comparison, confirming that shape distortion shifts the enclosed volume distribution beyond naive radial estimates.

12.3 Experiment 3: Phase delay and group delay are not interchangeable

We evaluated a synthetic dispersive medium with a prescribed frequency-dependent phase index, stated for reproducibility: $n_p(\omega) = 1.401 + 0.006\,(\omega/\omega_0)^2$ (an explicitly anomalous model chosen so that $\omega\,\partial_\omega n_p > 0$), $\omega_0 = 2\pi \times 405\,\mathrm{THz}$, path length $L = 1\,\mathrm{m}$. Then $n_p(\omega_0) = 1.407$ exactly, and $\omega_0\,\partial_\omega n_p(\omega_0) = 2(0.006) = 0.012$, giving $n_g = n_p + \omega\,\partial_\omega n_p = 1.419$. For a homogeneous path, $T_p = n_p L/c$ and $T_g = n_g L/c$; using phase delay to correct a group-delay measurement introduces $(n_g - n_p)L/c \approx 4.0 \times 10^{-14}\,\mathrm{s} = 40\,\mathrm{fs}$. The synthetic configuration thus produced a $\approx 0.85\%$ relative timing discrepancy per fixed path length when defined as $(n_g - n_p)/n_g$, and $\approx 0.86\%$ when defined as $(n_g - n_p)/n_p$; the denominator is stated so the number is reproducible. This is not a numerical indexing error; it is a mismatch between the modeled observable and the measured one. The result reinforces the distinction between the phase dispersion relation, which generates the rays, and group delay, which supplies the relevant timing correction.

12.4 Experiment 4: Canonical chart ownership prevents address aliasing

We tested the article's three-chart address table with $(B_0, N_0) = (0, 512)$, $(B_1, N_1) = (512, 256)$, $(B_2, N_2) = (768, 256)$; the arena contains $N = 1024$ addresses. For each valid chart–rank pair we computed $E(k, r) = B_k + r$. Exhaustive enumeration produced 1,024 addresses, all unique, covering the complete integer interval $[0, 1024)$. Therefore, for this concrete table, the encoder is a bijection between valid chart–rank pairs and arena addresses, instantiating Theorem 6 for a finite case.

The experiment also illustrates why the chart identifier is essential: if local ranks are used as global addresses, cells belonging to different charts alias—in this table, ignoring the chart id yields 384 duplicate ranks. This exhaustive test establishes the bijection for the tested finite table. It does not constitute a general SPARK proof for arbitrary chart tables, nor does it validate the optical correctness of the chart partition. Evidence type for this experiment is (B) for the table; (C) is still missing for the general implementation.

12.5 Experiment 5: Uniform bins versus task-aware allocation—held-out

We compared alternative partitions under the same cell budget $K = 64$. The tested delay function $T_g(\Omega)$ varied slowly over one part of the angular domain and rapidly over another. A uniform partition allocated equal resolution to both angular regions; a dynamic-programming partition optimized contiguous angular cells to minimize $\mathcal{L}(\mathcal{P})$ on training data, subject to the chart-legality constraints (contiguity, one interval per branch, branch identity recoverable from the address). From 300,000 rays per seed under the deterministic SHA-256 80/20 split we used the first 200,000 of the $\approx$ 240,000 available training rays and all $\approx$ 60,000 held-out rays ($60{,}016 \pm 223$ across seeds;

60,170 at the reference seed). Exact held-out improvement for the chart-legal DP partition:

$$\frac{0.000423}{0.000242} = 1.748 \approx \mathbf{1.75}\times \tag{46}$$

reduction in within-cell distortion.

What the headline number does and does not mean. A single ratio invites over-reading, so we report three controls, all at released replication cost (12 seeds, $t_{11} = 2.201$; `code/scaling.py`).

First, the improvement is a property of the budget as much as of the method.

K (cells/branch)	uniform fs^2	DP fs^2	ratio	95% CI
16 (8)	6.406×10^{-3}	2.658×10^{-3}	2.410	$\pm$0.011
32 (16)	1.677×10^{-3}	8.195×10^{-4}	2.046	$\pm$0.007
64 (32)	4.231×10^{-4}	2.421×10^{-4}	1.748	$\pm$0.009
128 (64)	1.059×10^{-4}	6.689×10^{-5}	1.583	$\pm$0.005
256 (128)	2.649×10^{-5}	1.757×10^{-5}	1.508	$\pm$0.007

The advantage decays with budget and converges towards the generic value in the next paragraph. Any single-budget figure is therefore a statement about that budget.

Second, most of the gain is ordinary optimal nonuniform quantization. At $K = 64$ (32 cells per branch) the same code applied to other delay laws gives:

delay law	interpretation	ratio	95% CI
$A_0\theta$	affine in Ω	0.997	$\pm$0.002
$A_0 \sin 2\theta$	one smooth regime, no split	1.504	$\pm$0.006
$A_0 A \sin 4\theta$	the released two-regime law	1.748	$\pm$0.009
$A_0 \sin 8\theta$	one fast regime	1.889	$\pm$0.004

The affine row is the control that matters most: uniform bins are optimal when the observable is affine, so the estimator is not simply favouring itself, and 0.997$\times$ is the expected null. The smooth-sinusoid row shows that a delay law with *no* slow/rapid structure at all already delivers 1.50$\times$ at the same budget. The two-regime structure of our released model therefore adds to a generic effect rather than causing it, and the honest summary is that this experiment confirms the standard result — optimal nonuniform quantization beats uniform binning on any curved observable [9–11] — within an architecture that must also carry optical identity.

Third, the sample is not doing the work it appears to do. The released delay law is known in closed form, so the identical dynamic program can be run on an analytic quadrature grid with zero sampled rays. It reaches 2.4192×10^{-4} fs^2 against the sample-estimated 2.4208×10^{-4} fs^2, a relative gap of 0.07%, with cell boundaries agreeing to within 1–3 grid points out of 1024. The partition is a function of the delay law; the 200,000 rays estimate it and add a little noise. We therefore describe this procedure as partition optimization *estimated from samples*, not as learning from data, and note that for a calibrated synthetic benchmark the analytic route is both cheaper and marginally better.

Chart legality is an address-structure cost, not a moral one. The sharpest way to price the chart-legal constraint is to compute what it forbids. Optimizing the scalar observable alone — the classical optimal scalar quantizer of T_g (Max 1960 [10]; Lloyd 1982 [9]; Gray–Neuhoff 1998

[11]), precisely the prior work this paper cites — with 32 cells reaches $4.0019 \times 10^{-5}\,\mathrm{fs}^2$, a 10.6× improvement over uniform binning rather than 1.75×. That solution is not incorrect; it is unrepresentable. Its cells are contiguous in T_g but not in Ω, and scanning Ω they fragment into 91 runs, so realizing them requires $\approx 2.8\times$ the chart-address structure of the 32 contiguous angular cells we are permitted. The earlier framing of this experiment — an inadmissible competitor that gains 1.67× by merging opposite branches — understated the effect, because it compared a chart-legal optimum against a weak heuristic rather than against the true scalar optimum.

This yields the result we consider the paper's sharpest statement about representation cost. The scalar optimum is dramatically better and the chart-legal optimum is dramatically worse, and the difference is not a moral judgement but a measurable requirement for address structure. Consequently the design objective of §11 is correctly specified: one cannot price a representation by scalar reconstruction error alone, because topology, identity, address count, calibration, and distortion are coupled resources. Theorem 11 prices one analytic instance of that coupling; the 91/32 measurement above prices a geometric instance of the same coupling, and the two are independent routes to the same conclusion.

The released reference generator reproduces both headline distortions (4.230×10^{-4} and $2.423 \times 10^{-4}\,\mathrm{fs}^2$) and the 1.746 ± 0.005 ratio over 30 seeds, byte-identically; the scaling, control, and Lloyd–Max tables above are regenerated by `code/scaling.py`.

The result illustrates why a purely geometric refinement heuristic can fail: a small ray-coordinate Jacobian may indicate a caustic, but it does not by itself quantify the benefit of adding a memory cell. The relevant quantity is the reduction in expected distortion achieved by refinement,

$$\boxed{\Delta\mathcal{L}(C) = \mathcal{L}(\mathcal{P}) - \mathcal{L}(\mathcal{P} \text{ with } C \text{ refined}),} \tag{47}$$

and for a fixed memory budget, refinement should prioritize cells with the greatest expected reduction in timing distortion per unit of additional storage, subject to coverage and hardware constraints.

Repeated-seed statistics and released reproduction artifacts. The original Experiment 5 run was not replicated, so no uncertainty was originally reported. We therefore calibrated and released a reference generator (`code/generate.py`) whose two calibration constants are documented: an amplitude fixing the uniform-bin distortion at $4.23 \times 10^{-4}\,\mathrm{fs}^2$ and a heterogeneity parameter fixing the DP ratio at the documented value. Over 30 independent sampling seeds (300k rays each under a deterministic SHA-256 80/20 split, from which the first 200k training rays and all $\approx$ 60k held-out rays are used; exact DP on a 1024-point boundary grid; branch identity enforced by $K/2$ addresses per branch) the mean held-out distortions were

$$\overline{D}_{\text{uniform}} = 4.230 \times 10^{-4} \pm 9.0 \times 10^{-7}, \qquad \overline{D}_{\text{DP}} = 2.423 \times 10^{-4} \pm 4.2 \times 10^{-7} \tag{48}$$

(95% CI, t_{29}), giving a per-seed improvement ratio $\mathbf{1.746 \pm 0.005}$ (95% CI). The ratio is stable across seeds; the absolute distortion scale is set by the calibrated amplitude, and the confidence interval quantifies split and sampling variability under the released model, not model-selection uncertainty. All per-seed values are in `code/results_exp5_seeds.csv`.

Figure 2 (numerical summary). Held-out within-cell distortion at equal cell budget $K = 64$, on the released calibrated model: DP improves 1.75× over uniform angular binning, while optimizing the scalar observable alone would improve 10.6× but requires 91 angular runs instead of 32.

One clarification about scope. The released delay law is branch-independent and is not π-symmetric: $T_g(\theta) \neq T_g(\pi - \theta)$ for all $\theta \neq \pi/2$. This generator therefore *cannot* exhibit a $\theta \leftrightarrow \pi - \theta$ branch collision, and any contiguous angular partition preserves branch identity trivially on it. Branch identity is accordingly a theoretical architectural constraint throughout this paper (Sec. 6, Theorem 11), not a quantity Experiment 5 measures; what Experiment 5 measures is chart-legal contiguous quantization of a scalar observable. We removed an earlier draft claim that a released transform-based partition merged $\theta = 0.35\pi$ with $\theta = 0.65\pi$ because they share a delay value, as the released model does not support it.

12.6 Calibration allocation at large and small M

Using the chart-legal DP bins of the previous subsection, with p_k and v_k estimated from training data and the continuous relaxation of Proposition 8: the optimal allocation offers negligible improvement when calibration is abundant but becomes consequential when samples are scarce, because the calibration term $p_k v_k / m_k$ grows as $1/m_k$ while the within-cell term stays fixed. At $M = 10^6$, optimal $\{m_k\}$ is nearly uniform; at $M = 256$, optimal allocations deviate strongly from uniformity, showing the continuous relaxation breaks down and integer allocation is required. Finite-calibration simulation for the binned lifted estimator with floor variance $0.117\,\mathrm{fs}^2$ (calibration-term inflation v_k/m_k, $v_k \approx 0.117$): $m = 1$ gives $+100\%$ (variance $0.234\,\mathrm{fs}^2$), $m = 2$ gives $+50\%$, $m = 4$ gives $+25\%$, and $m = 16$ gives $+6.25\%$; article.md's $+16.6\%$ corresponds to $m = 6$. A cell cheap in silicon may be expensive to calibrate.

13 Discussion: the memory problem was never just about memory

We began with five logical dimensions and a contiguous memory requirement. Ordinary row-major indexing solved the dimensionality problem but ignored the domain's geometry. A compact spatial-rank lookup solved wasted allocation but did not establish the locality and sampling properties needed by the optical application. Progressive ellipticity forced us to examine the geometry itself. Fermat's principle introduced optical travel time; the eikonal equation produced equal-phase wavefronts; the Cauchy problem and Hamiltonian characteristics described how those wavefronts propagate; caustics showed why spatial ray coordinates can lose invertibility; the fixed-frequency optical phase space explained the natural appearance of three positional and two directional degrees of freedom; a multi-chart partition provided a route to contiguous, collision-free digital addresses; Ada and SPARK gave us tools to express and potentially verify the discrete representation's invariants; and the phase-versus-group-delay distinction forced the separation between the manifold, generated by the phase dispersion relation, and group delay, a different observable computed along its rays. The memory layout must choose which functional it organizes by: for a pulse-arrival observable, group delay is a natural leading-order choice, but that does not license rebuilding the manifold from the group index. State space and organizing coordinate are separate decisions.

The atomic-clock application explains why all this matters: the system must not confuse geometric propagation effects with genuine timing fluctuations. There is one last mathematical distinction worth appreciating. Hamiltonian evolution preserves symplectic structure in full phase space; under suitable conditions, Liouville's theorem [29] tells us that phase-space volume is preserved by the flow. But preservation of full phase-space volume does not mean spatial volume is preserved: a ray bundle can focus dramatically in position space while spreading in momentum directions. That is why caustics are compatible with well-behaved Hamiltonian evolution, and it is why a represen-

tation that keeps only spatial coordinates can become inadequate. The mathematics was quietly telling us where to put the missing information: not in another arbitrary array—in the directions of the rays.

13.1 Interpretation of all five experiments

The five experiments expose five different failure modes at different layers of the system; fixing one does not automatically fix the others.

- **Spatial projection** can manufacture apparent timing fluctuations by merging physically distinct ray branches (Exp. 1).
- **Uniform logical sampling** can misrepresent physical resolution in an anisotropic domain (Exp. 2).
- A **phase-versus-group-delay mismatch** can introduce systematic timing error even when the data structure itself is correct (Exp. 3).
- Omitting **canonical chart ownership** can produce actual memory-address collisions (Exp. 4).
- **Unconstrained task-aware optimization** can improve a single timing number while destroying branch identity (Exp. 5).

The central experimental conclusion is therefore not that a five-dimensional array is inherently superior to a three-dimensional array. It is that preserving the optical state variables required by the measurement prevents information loss that a spatial-only representation cannot recover after projection. The experiments establish this result in controlled synthetic settings; physical validation, broadband propagation modeling, and machine-checked verification remain separate obligations.

14 Five theoretical extensions

The five results in this section are deductions from the model of the preceding sections, with each additional assumption stated explicitly. The first limits the five-dimensional representation itself.

14.1 Closed-ray obstruction: even exact position and direction may not determine accumulated delay

Suppose an admitted ray returns to the same position *and* direction after a complete circuit, with positive group index along the circuit.

Theorem 9 (Closed-ray obstruction)**.** *Accumulated group delay cannot be represented by a single-valued function of the instantaneous five-dimensional optical state on any domain that includes repeated circuits.*

Proof. Let $q(s)$ denote the position–direction state, with $q(P) = q(0)$ after one circuit of period P. If accumulated delay were a state function $G(q)$, returning to the same state would imply $G(q(P)) - G(q(0)) = 0$. But propagation around the circuit accumulates $\Delta T_g = \oint_\gamma \frac{n_g}{c}\, d\ell > 0$. The same difference would have to be both zero and positive—a contradiction. □

This is a continuous-state obstruction: increasing angular or spatial resolution cannot fix it. The caustic argument of §5 distinguishes rays with different directions; this result identifies a further ambiguity: two visits can have identical position and direction but different travel times. The five-dimensional representation can identify the instantaneous ray state while failing to identify the measurement's history, so a recirculating instrument needs additional information such as a lap

counter, launch-event identity, or accumulated delay. Single-pass acquisition avoids this obstruction by selecting one traversal from an entry surface.

14.2 Hidden branches can manufacture an entire clock-like noise spectrum

The paper proves that unresolved geometry adds variance; a stronger result concerns its *temporal structure*. Suppose two branches share one stored address with delays $T_g(Z_t) = T_0 + aZ_t$, $Z_t \in \{-1, +1\}$, where the branch sequence is a stationary symmetric Markov chain with $\Pr(Z_{t+1} = Z_t) = \frac{1+\rho}{2}$, $0 < \rho < 1$. Assume a perfectly stable clock and independent white detector noise ε_t of variance σ^2. After subtracting the merged-cell correction T_0, the residual is $R_t = aZ_t + \varepsilon_t$.

Theorem 10 (Clock-like spectrum)**.** *The residual has autocovariance and discrete-frequency power spectrum*

$$\boxed{\mathrm{Cov}(R_t, R_{t+h}) = a^2\rho^{|h|} + \sigma^2\mathbf{1}_{\{h=0\}}, \qquad S_R(\Omega) = \sigma^2 + a^2\frac{1-\rho^2}{1-2\rho\cos\Omega+\rho^2}.} \tag{49}$$

Proof. The transition rule gives $\mathbb{E}[Z_{t+h} \mid Z_t] = \rho^h Z_t$. Since $\mathbb{E}[Z_t] = 0$ and $Z_t^2 = 1$, the covariance follows directly; Fourier transforming the geometric sequence gives the spectrum. □

At zero frequency the geometric contribution is $S_{\mathrm{geom}}(0) = a^2\frac{1+\rho}{1-\rho}$: as branch persistence approaches one, substantial low-frequency power appears even though the clock is perfect. There is also an identifiability consequence: without additional restrictions or branch observations, the explanations (i) stable clock with fluctuating unresolved branch delay aZ_t and (ii) constant propagation delay with clock fluctuation aZ_t produce exactly the same recorded residual process. Temporal correlation alone cannot distinguish hidden optical geometry from clock fluctuations; validation should therefore test autocorrelation and spectra after branch correction, alongside the residual variance already considered, using the metrological machinery of Allan statistics [7].

14.3 Preserving branch identity has an exact distortion cost

Consider a controlled model: J equally probable branches, delay T uniformly distributed over an interval of width W on every branch, branch identity independent of delay, K total addresses with K divisible by J, one reconstructed delay per address, and branch identity recoverable from the address. For a uniform scalar, the optimal k-cell squared-error distortion is $W^2/(12k^2)$, obtained by equal intervals and midpoint reconstruction [9, 11].

Theorem 11 (Branch-preservation cost)**.** *Subject only to exact branch preservation,* $D^\star_{\mathrm{preserve}}(K) = \frac{J^2W^2}{12K^2}$*; if branch identity may be discarded,* $D^\star_{\mathrm{discard}}(K) = \frac{W^2}{12K^2}$*; consequently*

$$\boxed{\frac{D^\star_{\mathrm{preserve}}}{D^\star_{\mathrm{discard}}} = J^2.} \tag{50}$$

Proof. Exact branch recovery requires disjoint address sets. Let branch b receive K_b addresses with $\sum_{b=1}^{J} K_b = K$; its conditional distortion is at least $W^2/(12K_b^2)$, so $D \geq \frac{W^2}{12J}\sum_{b=1}^{J}\frac{1}{K_b^2}$. Convexity of x^{-2} makes equal allocation $K_b = K/J$ optimal; substitution proves the result, and equal delay intervals attain it. □

In this model, preserving four independent branch identities costs a factor of 16 in MSE—or 4 in RMS error—at the same total address capacity. An inadmissible method can therefore win a timing benchmark by a large factor purely because it discards required information: the counterexample of Experiment 5 has an exact theoretical counterpart. The budget must count the combined identity; storing branch labels separately consumes additional bits even when correction tables can be shared. This places the branch-constrained design inside the functional-quantization conversation [12], where quantizers minimize the error of a computed function rather than a raw reconstruction, and into the broader quantization theory [8, 11].

14.4 Refinement has a calibration threshold—and useful resolution grows only as $M^{1/3}$

For partition $\mathcal{P}$ define $p_k = \Pr(Q \in C_k)$ and $v_k = \mathrm{Var}(T_g \mid C_k)$; let each correction be estimated from m_k independent calibration observations, each carrying additional independent noise of variance ν^2, and write $D(\mathcal{P}) = \sum_k p_k v_k$ and $S(\mathcal{P}) = \sum_k \sqrt{p_k(v_k + \nu^2)}$.

Theorem 12 (Calibration threshold)**.** *Under a total calibration budget $\sum_k m_k = M$, the continuous allocation optimum is*

$$\boxed{R^\star(\mathcal{P}) = \sigma^2 + D(\mathcal{P}) + \frac{S(\mathcal{P})^2}{M},} \tag{51}$$

where σ^2 is evaluation detector-noise variance.

Proof. Independent calibration and evaluation give $R = \sigma^2 + D + \sum_k \frac{p_k(v_k+\nu^2)}{m_k}$. Cauchy–Schwarz yields $\big(\sum_k \sqrt{p_k(v_k+\nu^2)}\big)^2 \le \big(\sum_k \frac{p_k(v_k+\nu^2)}{m_k}\big)\big(\sum_k m_k\big)$, with equality for $m_k = M\,\frac{\sqrt{p_k(v_k+\nu^2)}}{S(\mathcal{P})}$, the continuous relaxation of Proposition 8 extended by ν^2; actual sample counts require integer allocation. □

Consequently, a refinement $\mathcal{P} \to \mathcal{P}'$ improves the relaxed total error *if and only if*

$$\boxed{D(\mathcal{P}) - D(\mathcal{P}') > \frac{S(\mathcal{P}')^2 - S(\mathcal{P})^2}{M}.} \tag{52}$$

Reducing geometric distortion alone is insufficient. For an exact counterexample, let the true delay be constant everywhere: with K equally probable cells, separate sample-mean calibration gives $R - \sigma^2 = K\nu^2/M$—every additional cell increases error while providing no geometric benefit.

A useful scaling law follows for an equal-probability design family with within-cell variance A/K^2. Equal calibration allocation gives $R(K) - \sigma^2 = \frac{A}{K^2} + \frac{A}{MK} + \frac{\nu^2 K}{M}$; for $K \ll M$ and $\nu^2 > 0$, balancing the dominant terms gives

$$\boxed{K_{\mathrm{opt}} \approx \left(\frac{2AM}{\nu^2}\right)^{1/3},} \tag{53}$$

subject to memory and sampling feasibility. Doubling useful resolution therefore requires approximately eight times the calibration budget; the $M^{-2/3}$ error scaling connects to classical nonparametric estimation rates [13]. The refinement rule of §6 should thus optimize expected *measurement-error* reduction after calibration, rather than geometric distortion reduction alone.

14.5 Ray-path sensitivity contributes to dispersion through an inverse Hessian

This strengthens the phase-versus-group-delay discussion of §9. Parameterize admissible fixed-endpoint paths by a finite vector z with redundant coordinates removed, and define the accumulated phase

$$\Phi(z,\omega) = \frac{\omega}{c}\int_{\gamma_z} n_p(\mathbf{x},\omega)\, d\ell. \tag{54}$$

A smooth physical branch satisfies $\nabla_z \Phi(z_\star(\omega),\omega) = 0$; assume the path Hessian $H = \Phi_{zz}$ is invertible. For the stationary-path phase $\phi(\omega) = \Phi(z_\star(\omega),\omega)$, group delay is $\phi'(\omega)$ and group-delay dispersion is $\phi''(\omega)$ [14].

Theorem 13 (Dispersion sensitivity)**.** *On the stationary branch,*

$$\boxed{T_g = \phi'(\omega) = \Phi_\omega = \int_{\gamma_{z_\star}} \frac{n_g}{c}\, d\ell, \qquad \frac{dT_g}{d\omega} = \Phi_{\omega\omega} - \Phi_{\omega z} H^{-1} \Phi_{z\omega}.} \tag{55}$$

Proof. Differentiating stationarity: $H z_\star'(\omega) + \Phi_{z\omega} = 0$, so $z_\star' = -H^{-1}\Phi_{z\omega}$. The first derivative of the stationary-path phase is $\phi' = \Phi_\omega + \Phi_z z_\star' = \Phi_\omega$ because $\Phi_z = 0$ on the stationary branch. Differentiating once more and substituting $z_\star'$ proves the dispersion formula. □

The important asymmetry: path variation cancels from the *first* phase derivative but returns in the *second*. If H has a small eigenvalue λ with unit eigenvector e, the path contribution includes $-|e^\top \Phi_{z\omega}|^2/\lambda$, which can become large when frequency couples to a weakly constrained path deformation. The formula applies while the branch remains nondegenerate; at singularity, a different local treatment is needed. Assessing bandwidth using material dispersion alone can therefore miss sensitivity caused by frequency-dependent ray bending. For a smooth branch over $|\omega - \omega_0| \le h$, the mean-value theorem supplies the practical bound

$$\boxed{|T_g(\omega) - T_g(\omega_0)| \le h \sup_{\text{band}} \left|\Phi_{\omega\omega} - \Phi_{\omega z} H^{-1} \Phi_{z\omega}\right|,} \tag{56}$$

and requiring this to stay below the timing tolerance gives a concrete criterion for when one frequency-specific calibration table is adequate and when the frequency partition of §9 needs refinement.

14.6 What the extensions change in the design

- **Extension 1:** recirculating instruments need history variables (lap counter, launch identity, or accumulated delay) beyond the five-dimensional state.
- **Extension 2:** validation must test autocorrelation and spectra after branch correction, not only residual variance; clock-like spectra can be geometric artifacts.
- **Extension 3:** the benchmark budget must count branch identity exactly; J branches cost J^2 in MSE relative to identity-discarding compression.
- **Extension 4:** refinement must clear the calibration threshold of Theorem 12; the paper's refinement rule should target post-calibration error, and constant-delay cells are strictly harmful.
- **Extension 5:** bandwidth assessment needs the inverse-Hessian dispersion sensitivity, not material dispersion alone.

15 Limits and unproved obligations

By Thursday afternoon, the prototype has produced: a reproducible optical model, a finite phase-space partition, a contiguous Ada memory arena, an explicit set of address invariants, and a test harness designed to send rays through a caustic. It knows the difference between two rays meeting at the same place; it knows that outer shells do not behave like inner shells; it knows that the phase index and the group index are not interchangeable; and it knows that a successful memory lookup is not proof of a successful physical measurement. It has not discovered a new fundamental clock, the SoC is not fabricated, and nobody has proved that nature contains a subtler clock beneath the atomic clock.

Three things remain unsettled, and we state them as specific, testable claims rather than as failures:

(1) **Implementation verification scope.** The released Ada/SPARK specification is machine-checked by GNATprove (47/47 proof obligations discharged, 0 unproved, for the released three-chart encoder instance and its contract structure; Appendix A), which closes Tier 1 for the released chart table. The chart table is a three-entry literal, so injectivity holds there by construction; the table consistency predicate is not used as any subprogram's precondition, the ghost `Global_Rank` entry point is an unimplemented stub, and the exhaustive bijection test is executed rather than verified. This does not cover the geometry preprocessor, the interrupt-time chart-table update path, or arbitrary chart tables; those remain outside the transcript's scope and are stated as such.

(2) **Geometric preprocessing quality.** The quality of the geometric preprocessing—quantified cell coverage and approximation error near caustics and deformed boundaries—is a numerical question the four-day schedule does not close (Tier 2, §10).

(3) **Metrological benefit.** Whether this architecture improves a real femtosecond measurement over a simpler baseline is a scientific question no data structure can answer on its own (Tier 3).

These are the criteria on which the prototype must eventually be judged. We close with the paper's conclusion, restated once more: the optimal representation is not the one that stores the fewest coordinates or minimizes a single timing error; it is the least expensive representation that preserves every distinction the instrument may need to make. The intern started with an array and ended with a constrained information-theoretic instrument-design problem.

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A Ada/SPARK specification and GNATprove transcript

A.1 Specification (`chart_encoder.ads`)

```
--  Chart-aware contiguous memory encoder for discretized optical phase space.
--  Companion artifact to "Chart-Aware Contiguous Memory for Discretized Optical
--  Phase Space" (Sethurathienam Iyer, 2026).
```

```
--
--  Design intent (paper Sections 7-8):
--    * E(k, r) = B(k) + r   with consecutive intervals B(0) = 0,
--      B(k+1) = B(k) + N(k).
--    * Range bound        : every address produced is < Arena_Size.
--    * Injectivity        : distinct (chart, rank) pairs yield distinct addresses
--      (relational property, proved via ghost lemma in body).
--    * Coverage/bijectivity follows from consecutiveness of the intervals.
--
--  The chart table below is the concrete table used in Experiment 4:
--  (B0, N0) = (0, 512), (B1, N1) = (512, 256), (B2, N2) = (768, 256).

package Chart_Encoder with SPARK_Mode is

   Chart_Count : constant := 3;
   Arena_Size  : constant := 1024;

   type Chart_Id  is new Natural range 0 .. Chart_Count - 1;
   type Rank      is new Natural range 0 .. Arena_Size - 1;
   type Address   is new Natural range 0 .. Arena_Size - 1;

   --  Chart table (immutable at acquisition time; versioned per batch).
   Base     : constant array (Chart_Id) of Natural := (0, 512, 768);
   Cells    : constant array (Chart_Id) of Natural := (512, 256, 256);

   --  Contract on the raw table: intervals are consecutive and in bounds.
   Table_Is_Consistent : constant Boolean :=
     Base (0) = 0
     and then (for all J in Chart_Id =>
                 Base (J) + Cells (J) <= Arena_Size)
     and then (for all J in Chart_Id'First .. Chart_Id'Last - 1 =>
                 Base (Chart_Id'Succ (J)) = Base (J) + Cells (J));

   --  Total number of cells across all charts.
   Cell_Total : constant := 512 + 256 + 256;  --  = Arena_Size

   type Valid_Rank is new Natural range 0 .. Cell_Total - 1;

   --  The map from linear ranks within a chart to addresses is injective
   --  globally: this predicate expresses that (K, R) corresponds to a
   --  unique global rank, hence a unique address.
   function Global_Rank (K : Chart_Id; R : Valid_Rank) return Natural with
     Pre     => Natural (R) < Cells (K),
     Global    => null,
     Import,
     Ghost;
   --  Encoder: the ONLY address-producing primitive.
   --  The rank must lie within the chart's own cell count; the range bound
```

```
   --  then follows from the consecutive-interval table invariant.
   function Encode (K : Chart_Id; R : Valid_Rank) return Address with
     Pre  => Natural (R) < Cells (K)
               and then Base (K) + Natural (R) < Arena_Size,
     Post => Natural (Encode'Result) = Base (K) + Natural (R);

   --  Decoder: reconstructs the canonical cell identity from an address.
   --  Establishes Tier 1 representational correctness: round trip.
   procedure Decode
     (A : Address;
      K : out Chart_Id;
      R : out Valid_Rank) with
     Post => Encode (K, R) = A;

end Chart_Encoder;
```

The body adds a ghost lemma `Injectivity_Lemma(K1,R1,K2,R2)` whose precondition includes the pairwise-disjoint consecutive-interval table invariant and whose postcondition concludes $K_1 = K_2 \wedge R_1 = R_2$.

A.2 Transcript (GNATprove 14.1.1; cvc5, Alt-Ergo, Z3)

```
Summary of SPARK analysis
-------------------------------------------------------------------------------
SPARK Analysis results        Total      Flow      Provers   Justified   Unproved
-------------------------------------------------------------------------------
Initialization                    2         2            .           .          .
Run-time Checks                  29         .    29 (CVC5)           .          .
Assertions                        9         .     9 (CVC5)           .          .
Functional Contracts              6         .     6 (CVC5)           .          .
Termination                       1         1            .           .          .
-------------------------------------------------------------------------------
Total                            47    3 (6%)     44 (94%)           .          .

  Chart_Encoder          flow analyzed and proved (4 checks)
  Chart_Encoder.Encode   flow analyzed and proved (13 checks)
  Chart_Encoder.Decode   flow analyzed and proved (6 checks)
  Chart_Encoder.Injectivity_Lemma  analyzed and proved (21 checks)
```

The exhaustive runtime test (`test_encoder.adb`) additionally enumerates all 1024 chart–rank pairs, confirms no duplicate addresses, checks that the union covers $[0, 1024)$, and verifies the decode round trip. Both the specification and the transcript are released with the paper (`code/chart_encoder.*`, `code/gnatprove-transcript.txt`). Scope: the proof applies to this chart table and contract structure; it does not cover the geometry preprocessor or the interrupt-time update path (§15).

B Contact geometry of the fixed-frequency manifold

The fixed-frequency optical manifold $\mathcal{M}_{\omega_0}$ is a regular hypersurface in the cotangent bundle of three-dimensional space. The ambient cotangent bundle has a canonical one-form $\lambda = \mathbf{p} \cdot d\mathbf{x}$; restrict

it to the hypersurface, $\alpha = \lambda|_{\mathcal{M}_{\omega_0}}$. For a smooth positive isotropic refractive index, this defines a *contact form* [6]: its kernel is a maximally non-integrable four-dimensional plane field on the five-dimensional manifold. The associated Reeb vector field is tangent to the optical characteristic flow, its trajectories coinciding with the phase-generated ray trajectories up to reparameterization. The full six-dimensional phase space is symplectic, while its fixed-frequency optical hypersurface carries a natural contact structure; the relationship can be understood through symplectization, the positive radial extension of the contact hypersurface.

This explains why five dimensions arise so naturally: they are not merely the result of deleting one momentum coordinate from a six-dimensional array; they are the dimension of a contact manifold whose distinguished flow describes optical rays. None of this is necessary for the four-day prototype. It suggests a deeper research direction—whether a discrete optical state representation can preserve useful contact-geometric invariants with controlled approximation error—which is far more ambitious than whether an Ada array can store five indices. It also explains why chart-aware addressing is geometrically natural, even though contact geometry alone does not prescribe a unique or optimal memory layout.

C Phase 2: closed rays, a dynamical zeta, quantum graphs, and an obstruction

Appendix B put a contact form $\alpha = \mathbf{p} \cdot d\mathbf{x}$ on the five-dimensional fixed-frequency manifold, with the Reeb flow recovering the optical characteristic flow up to reparameterization. Section 14 proved the closed-ray obstruction. This appendix follows the global closed-ray structure in the standard dynamical-systems sense. Nothing here is required for the four-day prototype. The aim is to mark precisely what the geometry already gives, what a small model can demonstrate numerically, and what fails.

C.1 Periods are orbit invariants

Let $\mathcal{P}$ be the set of primitive closed Reeb orbits: closed optical rays traversed once. Along $\gamma \in \mathcal{P}$ the Reeb condition $\alpha(R_\alpha) = 1$ identifies Reeb time with the contact action,

$$T_\gamma^{(p)} = \oint_\gamma \alpha = \oint_\gamma \frac{n_p(\mathbf{x}, \omega_0)}{c}\, d\ell, \tag{57}$$

and the group-delay observable supplies a second invariant on the same orbits, $G_\gamma = \oint_\gamma (n_g/c)\, d\ell$. These are not independent data. With the positive dispersion ratio $r = n_g/n_p$,

$$G_\gamma = \int_0^{T_\gamma^{(p)}} r\big(\phi^t q\big)\, dt, \tag{58}$$

so group delay is a positive time change of the phase period: the group-delay spectrum is the phase spectrum pushed through r, not a fresh object. Consequently a recirculating instrument carries two spectra of orbit invariants, $\{T_\gamma^{(p)}\}$ and $\{G_\gamma\}$, and the address map of §6 records neither at zero cost.

The closed-ray obstruction now reads as a statement about repetition. After m circuits, $q_m = q_0$ while $T_g(m) - T_g(0) = m\, G_\gamma$; the instantaneous five- dimensional state cannot distinguish $m = 1$ from $m = 2$, and the missing history variable *is* the repetition index of a periodic orbit. Appendix D gives the cohomological form of that statement.

C.2 A dynamical zeta function

For a reference time T_* define

$$\zeta_{\mathrm{opt}}(s) = \prod_{\gamma \in \mathcal{P}} \left(1 - e^{-sG_\gamma/T_*}\right)^{-1}, \tag{59}$$

with orbit norm $N_\gamma = \exp(G_\gamma/T_*)$ and logarithmic derivative $-\zeta'_{\mathrm{opt}}/\zeta_{\mathrm{opt}} = \sum_\gamma \sum_{m\geq 1} \log N_\gamma \, N_\gamma^{-ms}$, so that the periodic-orbit measure $\Theta_{\mathrm{opt}}(t) = \sum_\gamma \sum_{m\geq 1} G_\gamma \, \delta(t - mG_\gamma)$ has Laplace transform proportional to that derivative. Whether such a product converges in a right half-plane, and how far it continues analytically, is a property of the particular flow and must be established rather than assumed [30].

The Euler product, the repetition index m, and the log-weights assemble themselves exactly as in the Riemann Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$, whose logarithmic derivative is $-\zeta'/\zeta = \sum_p \sum_m (\log p) p^{-ms}$. This correspondence is the reason trace formulas resemble the explicit formula [31]. It is also why the correspondence is not by itself evidence of anything: every zeta function of this type has an Euler product over primitive objects with a repetition index.

C.3 An exact obstruction for finite additive length spectra

The cheapest arithmetic construction fails, and this is the one provable result of this appendix. Replace the medium by a finite optical graph with E edges of fixed optical lengths $\ell_1, \ldots, \ell_E$. Each closed orbit traverses edge i a non-negative integer number of times, so its period lies in the finite-rank additive module $\Lambda = \mathbb{Z}\ell_1 + \cdots + \mathbb{Z}\ell_E$. But the prime logarithms are linearly independent over $\mathbb{Q}$: if $\sum_j q_j \log p_j = 0$, clearing denominators gives $\sum_j m_j \log p_j = 0$ with $m_j \in \mathbb{Z}$, exponentiating gives $\prod_j p_j^{m_j} = 1$, and unique factorization forces every $m_j = 0$. Hence $\dim_{\mathbb{Q}} \mathrm{span}\{\log p\} = \infty$, which a rank-$E$ module cannot contain.

Theorem 14 (Finite-graph obstruction)**.** *No finite optical graph with additive edge-length weights realizes $C \log p$ as orbit periods for all primes. Any exact construction must leave the finite-metric-graph, additive-travel-time regime. [evidence type A]*

The escape routes are an infinite graph, a non-additive or state-dependent weight, or an operator whose spectrum encodes arithmetic without the classical lengths being $\log p$. We emphasise that Hilbert–Pólya never required racetracks of length $\log p$; it is the explicit optical construction of the primes that is closed here, not the operator route [32].

C.4 A three-loop model: counting, not equality

A weaker target survives, and a toy computation shows both why it looks impressive and why it is not. A bouquet of three loops a, b, c with lengths ℓ_a, ℓ_b, ℓ_c generates primitive orbits as primitive cyclic words, with periods forced to be $L(w) = n_a\ell_a + n_b\ell_b + n_c\ell_c$; no prime is supplied to the model. Fitting the three lengths to $\log p_1, \ldots, \log p_N$ (enumerating primitive necklaces to word-length 11, 25,486 orbits) gives

N	fitted (ℓ_a, ℓ_b, ℓ_c)	RMS rel. error	worst	entropy h
20	(0.7229, 1.1274, 1.6602)	2.13%	4.92%	0.99937
30	(0.7190, 1.1279, 1.6672)	1.80%	5.09%	0.99972
50	(0.71745, 1.13196, 1.66210)	1.48%	4.96%	1.00013

The fit is a shallow local optimum: independent restarts converge to $(0.7185, 1.1309, 1.6672)$ with RMS 1.49%, worst 4.96% and $h = 0.99903$, so the three lengths are determined to about three significant figures and the entropy to about $\pm 0.1\%$ [evidence type B].

For a suspension on a three-loop bouquet the topological entropy solves $e^{-h\ell_a} + e^{-h\ell_b} + e^{-h\ell_c} = 1$, and the fitted lengths land at $h \approx 1$. This has been read as an independent discovery that the system self-organises to the entropy required by the prime number theorem in logarithmic coordinates. It is not, and the reason is worth recording.

Proposition 15 ($h = 1$ is forced by the objective)**.** *Suppose a hyperbolic flow satisfies* $\pi_{\mathrm{orb}}(T) \sim e^{hT}/(hT)$ *and its primitive periods satisfy* $L_n \sim \log p_n$*. Then* $h = 1$*. [evidence type A]*

Proof. Evaluate the counting function at $T = \log p_n$. The left side is n; the right side is $e^{h \log p_n}/(h \log p_n) = p_n^h/(h \log p_n)$. Since $n \sim p_n/\log p_n$, agreement requires $p_n^h/h \sim p_n$, that is, $p_n^{h-1} \sim h$, which forces $h = 1$. □

So the optimiser was handed the answer: matching L_n to $\log p_n$ forces the counting law to agree asymptotically, and the counting law is what pins h. The departure from unity is the finite window and the fitting residual, not signal. The methodological point is the useful one: leading-order agreement is automatic and therefore worthless as evidence, so any advance must live in the fluctuation term.

C.5 The fluctuation test fails

At $T = 11.8$ the leading counts agree closely. With $e^T = 133{,}252$ one has $\pi(e^T) = 12{,}434$, $\mathrm{Li}(e^T) = 12{,}481$, and 12,496 primitive orbits below T. The orbit count sits within 0.12% of Li, exactly as the entropy argument predicts.

The residual structure does not agree. Comparing $R_{\mathrm{prime}}(T) = \pi(e^T) - \mathrm{Li}(e^T)$ with $R_{\mathrm{orb}}(T) = N_{\mathrm{orb}}(T) - \mathrm{Li}(e^T)$ after normalising the graph entropy to one gives correlations of approximately 0.24 (raw), 0.23 (normalised) and 0.15 (log-power spectral profiles) between the two residuals, and ≈ -0.02 between their first differences. The first-difference correlation is the decisive one: it is zero. The model reproduces the coarse growth and none of the arithmetic fluctuation [evidence type B].

This is the most useful result in the appendix, because it separates two phenomena that are easily conflated:

$$\boxed{\text{exponential orbit proliferation} \;\neq\; \text{Riemann-zero arithmetic structure.}} \tag{60}$$

Topological entropy controls the leading proliferation of periodic orbits. It does not produce the correlations encoded by the nontrivial zeros, and RH is precisely a statement about those zeros and their consequence for prime-counting error. A model that matches the prime number theorem's leading term while failing its fluctuations demonstrates that orbit proliferation is insufficient.

C.6 From orbits to a wave spectrum

Because optics is a wave phenomenon, the relevant question is whether the classical orbit structure leaves a fingerprint in a wave spectrum. Quantum graphs are the cleanest setting in which this can be answered exactly. A metric graph with a self-adjoint Laplacian has a secular equation $\det(I - U(k)) = 0$ combining bond propagation phases with vertex scattering amplitudes, and admits an exact periodic-orbit trace formula decomposing the density of states into a smooth Weyl term and an oscillatory orbit sum [30, 33].

For the three-loop bouquet with Kirchhoff–Neumann conditions at the single vertex, the secular equation reduces in closed form to

$$\sum_{i=a,b,c} \tan\left(\frac{k\ell_i}{2}\right) = 0. \tag{61}$$

Proposition 16 (Weyl counting for the three-loop bouquet)**.** *Let $L = \ell_a + \ell_b + \ell_c$. The eigenvalues of the Kirchhoff–Neumann Laplacian with $k \le K$, counted with multiplicity, satisfy $N(K) \sim LK/\pi$; the number of distinct wavenumbers satisfies $N_{\mathrm{dist}}(K) \sim LK/(2\pi)$, the factor two being the multiplicity of each root. [evidence type A]*

For $L = 3.51151$ and $K = 1200$ this predicts 1,341.30 eigenvalues, matching the enumerated 1,340; the corresponding distinct-wavenumber count is 670 against $LK/(2\pi) = 670.65$. We record the multiplicity explicitly because "number of resonances" is otherwise ambiguous by a factor of two.

Fourier transforming the spectral residual after subtracting the Weyl density produces peaks in the length domain. The first twenty prime-log targets matched detected peaks to 1.94% RMS, and primitive-orbit support for the first thirty to 1.74% RMS. A control set of 500 random entropy-one three-loop length assignments gave a median error of 5.66%, with none outperforming the tuned graph on the same criterion. These are optimised numerical agreements, not evidence of RH, and nearest-peak matching is sensitive to spectral crowding, degeneracy and resolution [evidence type B].

What the experiment does establish is narrower: the designed classical length structure leaves an observable fingerprint in the wave spectrum. That is the classical-to-wave correspondence a real optical experiment would need.

C.7 Thermal readout

Because the mode density enters the thermal spectrum, the same fingerprint should appear in a thermal readout. With occupation $\hbar\omega/(e^{\hbar\omega/k_BT} - 1)$ and a density of states split as $\rho = \bar{\rho}_{\mathrm{Weyl}} + \delta\rho_{\mathrm{orbit}}$, the thermal energy decomposes as $u(\omega, T) = u_{\mathrm{smooth}}(\omega, T) + \bar{E}(\omega, T)\,\delta\rho_{\mathrm{orbit}}(\omega)$ [34]. For an illustrative temperature and Gaussian broadening the residual carried $\approx 43\%$ of the smooth thermal component's RMS [evidence type B]. That percentage is not a physical prediction: it depends on the arbitrary broadening, temperature, window and coupling, and a real measurement would additionally require emissivity, loss, port coupling and detector response. The surviving conceptual point is that a thermal spectrum *can* inherit the periodic-orbit fingerprint of the underlying wave system. The structural parallel with the primes is the shared architecture—smooth counting law, discrete spectrum, and corrections carrying geometric or arithmetic information—not any identity of laws.

C.8 The Ihara–Bass detour and why the critical line does not transfer

Graphs carry a second zeta function whose product over primitive non-backtracking cycles converts to a finite determinant:[1]

$$Z_G(u)^{-1} = (1-u^2)^{m-n} \det\big(I - uA + u^2(D-I)\big) = \det(I - uB), \tag{62}$$

[1]The identity originates with Ihara [35] and was extended and generalised by Bass [36] and Hashimoto [37].

with A the adjacency matrix, D the degree matrix, and B the non-backtracking Hashimoto matrix. An infinite-looking product over primitive cycles becomes a finite-dimensional determinant, which is exactly the orbit-to-spectrum transformation one would want.

For a $(q+1)$-regular graph the Ihara zeta admits a Riemann-Hypothesis analogue: in the Ramanujan case the nontrivial poles lie on $|u| = q^{-1/2}$, and under $u = q^{-s}$ that circle becomes $\Re(s) = \frac{1}{2}$. It is essential that this is *not* a consequence of unitarity of a scattering matrix. The graph-theoretic critical-line statement depends on a specific spectral bound, closely tied to the Ramanujan property. The secular determinant of a self-adjoint quantum graph and the Ihara–Bass determinant are related members of one spectral-graph family but are not interchangeable formulas. A shared critical-line notation obtained by substituting $u = q^{-s}$ therefore does not transport any proof: what would be required is a genuine identity between the analytic continuations of the relevant spectral objects.

Finally, the exponent $\frac{1}{2}$ can be produced formally by multiplicative orbit weights: if each edge traversal carries magnitude $e^{-\ell_e/2}$ then an orbit has $|A_\gamma| = e^{-L_\gamma/2}$, and with $L_\gamma = \log p$ and a phase e^{-itL_γ} one recovers $p^{-1/2-it}$, the arithmetic exponent on the critical line. This is straightforward to engineer in an attenuating transfer model, and it is not compatible with arbitrary unitary quantum-graph scattering, since orbit amplitudes are products of scattering-matrix entries whose magnitudes cannot be assigned independently. More importantly, inserting $e^{-L/2}$ does not show that any determinant's zeros lie on the critical line; it reproduces one formal factor of the desired arithmetic weighting.

C.9 Scope: what this does not establish

A self-adjoint graph Laplacian has real eigenvalues, a periodic-orbit trace formula, and a dynamical zeta; its classical lengths can be tuned to resemble prime logarithms over a finite range. None of these facts identifies its spectral determinant with the completed Riemann zeta function. The Hilbert–Pólya requirement is not merely a self-adjoint operator with an interesting spectrum—such operators are abundant—but an exact correspondence with *all* nontrivial zeros, with correct multiplicities and no unwanted spectral objects. The explicit formula additionally carries precise prime-power weights, zero contributions, analytic continuation, functional-equation structure and correction terms; a generic graph trace formula acquires none of these from having many periodic orbits or unit entropy.

We therefore record the status of each element plainly. Established mathematics used here: the contact description of the optical characteristics (Appendix B), additivity of periods along closed trajectories, periodic-orbit Euler products, the prime-orbit counting law, exact quantum-graph trace formulas, and the Ihara–Bass identity. The exact finite-rank obstruction (Theorem 14) and the forcing of $h = 1$ (Proposition 15) are elementary but genuine. The three-loop fit, the spectral peak matching, the thermal residual and the fluctuation correlations are exploratory numerical results of a specific toy model [evidence type B]. The RH connection remains an analogy and a research direction, and the most valuable single outcome is the negative fluctuation result of the preceding subsections, which shows where a simple construction stops.

C.10 Next experiments and the return to memory

The next stage should stop optimising three loop lengths against more prime logarithms, which improves only a finite-window fit. The objective should move from orbit locations to spectral structure, optimising a trace-formula residual with terms for lengths, amplitudes, phases and spectrum, $\mathcal{J} = \lambda_L \mathcal{E}_{\text{lengths}} + \lambda_A \mathcal{E}_{\text{amplitudes}} + \lambda_\phi \mathcal{E}_{\text{phases}} + \lambda_S \mathcal{E}_{\text{spectrum}}$, subject to actual scattering constraints

rather than free orbit amplitudes, and penalising unmatched short orbits—matching a few desired trajectories is meaningless if the graph also produces a much larger population of undesired ones. Two further directions are more deeply founded than a small bouquet: hyperbolic surfaces, where Selberg's trace formula and zeta function give a rigorous geometry–spectrum correspondence through primitive closed geodesics [38], and a return to optics, constructing a refractive-index field with a controlled family of closed rays and studying its wave resonances.

Finally, the excursion returns to the original problem. At a caustic, distinct directions share a position; on a closed orbit, distinct traversal histories share an instantaneous state. Both failures have the same shape: the representation identifies objects the measurement must distinguish, and the remedy is not more numerical precision but the correct state distinctions. A primitive orbit and its repetitions are not interchangeable, and the distinction between $\gamma, \gamma^2, \gamma^3, \dots$ is exactly what the Euler product organises, just as primes and prime powers play distinct roles in the logarithmic derivative. The analogy does not make optical rays arithmetic objects. It does exhibit a shared principle: the identities and repetitions of fundamental objects matter, and a representation that collapses them discards information that cannot be recovered afterwards. In the instrument that loss appears as false timing noise; in a periodic-orbit calculation it destroys the orbit spectrum; in a spectral determinant incorrect multiplicities produce an entirely different analytic function.

D Cohomological form of the closed-ray obstruction

Theorem 9 is the optical instance of a statement about flows, cocycles and coboundaries, and stating it in that generality both sharpens it and identifies which part of the delay can be compressed. Let M be a smooth manifold, let $\phi^t : M \to M$ be the flow of a complete vector field X, and let $f : M \to (0, \infty)$ be a continuous delay accumulation rate with $D(x, T) = \int_0^T f(\phi^t x)\, dt$.

Theorem 17 (Coboundary obstruction)**.** *If ϕ has a periodic orbit γ of period $P > 0$, there is no globally defined C^1 function $u : M \to \mathbb{R}$ with $Xu = f$. [evidence type A]*

Proof. Suppose $Xu = f$ holds. Along γ, $\frac{d}{dt} u(\phi^t x) = (Xu)(\phi^t x) = f(\phi^t x)$. Integrating over one period gives $u(\phi^P x) - u(x) = \int_0^P f(\phi^t x)\, dt$. Since $\phi^P x = x$ the left side vanishes while the right side is strictly positive because $f > 0$. Hence $0 > 0$, a contradiction. □

Corollary 18. *Theorem 9 is the case in which M is the five-dimensional fixed-frequency optical manifold, ϕ the Reeb flow and $f = n_g/c$ the group-delay rate.*

The obstruction is therefore not an artefact of a particular data structure or a resolution limit; it is a statement that a strictly positive accumulation rate cannot be a gradient along the flow when a closed orbit exists. Two further observations convert it into a usable structure.

The first is that D is a dynamical cocycle. Additivity over concatenated intervals,

$$D(x, T + S) = D(x, T) + D\big(\phi^T x, S\big)\,, \tag{63}$$

is immediate from the definition, and is the defining property of an additive cocycle over $(\phi, \mathbb{R})$. A cocycle of the special form $D(x, T) = u(\phi^T x) - u(x)$ is a *coboundary*. By Theorem 17, positive delay around a closed ray is not a coboundary, so accumulated delay is a nontrivial cohomology class. The instantaneous state says where the ray is; the cocycle says what it has accumulated in getting there, and these are different kinds of information.

The second is that the obstruction is quantitative rather than binary, because the periodic-orbit integrals control which part of the cocycle is removable. This is the content of Livšic's theorem,

which for suitable hyperbolic systems states that vanishing integrals over all periodic orbits are sufficient for f to be a coboundary [39, 40]. Our observable fails that hypothesis immediately, but the theorem indicates the right decomposition. Fix a reference rate a and ask whether $f - a$ is a coboundary.

Proposition 19 (History/correction split)**.** *If $f - a = Xu$ for some globally defined u, then*

$$D(x, T) = aT + u(\phi^T x) - u(x), \tag{64}$$

and a single constant a is admissible if and only if $\oint_\gamma f\, dt = a\, P_\gamma$ for every periodic orbit γ, i.e. if and only if every closed orbit has the same mean delay rate a.

Thus the accumulated delay separates into an elapsed-history contribution aT and a state-dependent correction $u(\phi^T x) - u(x)$. The first is exactly the part that no instantaneous encoding can supply, and the second is compressible. This is a design principle for a delay-aware encoding: remove the coboundary component, retaining history information only for the non-coboundary part. It also bounds what the prototype of §6 could have achieved. A general flow need not admit a single lap counter, nor any finite-dimensional history variable valid for all trajectories, unless the periodic-orbit mean rates are degenerate in the sense of Proposition 19—which for a genuinely dispersive optical medium they are not.

We flag the limits of this appendix explicitly. Theorem 17 and Proposition 19 are proved under only smoothness and positivity. The Livšic route to a full characterisation of the removable component requires stated hyperbolicity and regularity hypotheses that we have not verified for the optical flow, and a memory-minimality theorem would require the encoder and reconstruction requirements to be formalised, which the prototype does not do.

E Observable-sufficient representation: the unifying criterion

The preceding sections are ordinarily read as separate techniques: a geometric lifting (§5), an encoder (§7), a rate–distortion allocation (§11), and a history obstruction (§14). This appendix shows that they are one statement, and that the failure modes and the costs are all instances of it. The organising notion is a criterion on the encoder.

E.1 The criterion

Let $\mathcal{H}$ be a set of admissible physical histories, $\mathcal{A}$ a memory representation, $\mathcal{Y}$ a measurement space, $E : \mathcal{H} \to \mathcal{A}$ an encoder and $Y : \mathcal{H} \to \mathcal{Y}$ the observable the instrument must report. Here a *history* H is the datum a real acquisition records—a trajectory segment with its initial state, its accumulated propagation, and any branch label—of which the instantaneous five-dimensional state q is only a summary.

Theorem 20 (Exact observable factorization)**.** *There exists a decoder $g : E(\mathcal{H}) \to \mathcal{Y}$ with $Y = g \circ E$ if and only if*

$$\boxed{E(H_1) = E(H_2) \implies Y(H_1) = Y(H_2)} \tag{65}$$

for all $H_1, H_2 \in \mathcal{H}$. [evidence type A]

Proof. If $Y = g \circ E$ and $E(H_1) = E(H_2)$, then $Y(H_1) = g(E(H_1)) = g(E(H_2)) = Y(H_2)$. Conversely, if the implication holds, define $g(a) = Y(H)$ for any H with $E(H) = a$; the hypothesis makes g well defined on $E(\mathcal{H})$, and $Y = g \circ E$ by construction. □

Table 1: Three aliasing modes, one criterion

Limitation	Violated condition	Response
Spatial aliasing	E_{3D} merges histories with Y different	lift to $\mathcal{M}_{\omega_0}$
Quantization aliasing	finite cells merge distinguishable states	constrained allocation
Historical aliasing	cocycle does not factor through q	retain history invariant

We call E *observable sufficient* for Y when the hypothesis of Theorem 20 holds. The theorem is elementary—it is the universal property of factoring a map through a quotient—but it locates the exact boundary of admissible compression: an encoder may merge two histories precisely when the instrument cannot distinguish them in the quantity it must report, and not one history more. For several required outputs one takes $Y = (T_g, B, Q, \dots)$ with B the ray-branch identity and Q any further mandatory invariant, which converts physical-identity preservation from an informal desideratum into a reconstruction contract.

E.2 The failure modes are instances

Spatial aliasing (caustic). At a caustic two histories share a position, so $E_{3D}(H_1) = E_{3D}(H_2)$, while differing directions give different delays, so $Y(H_1) \neq Y(H_2)$: the three-dimensional encoder fails Theorem 20. The residual this induces is exactly the $\mathrm{Var}_U(T_g)$ of Experiment 1, which the instrument reports as timing noise. Lifting to $\mathcal{M}_{\omega_0}$ restores sufficiency for single-pass geometry.

Historical aliasing (closed orbit). On a closed trajectory $q(P) = q(0)$, so even $E_{5D}(H_1) = E_{5D}(H_2)$, while $Y(H_2) - Y(H_1) = \oint_\gamma (n_g/c)\, d\ell > 0$. The five-dimensional encoder fails the criterion, and Theorem 17 explains why no refinement of it can succeed: the delay is a nontrivial cocycle, so the required decoder does not exist for any encoding that factors through the instantaneous state. Sufficiency requires extending E by a history invariant.

Finite resolution. Distinct states fall in the same cell, so the encoder merges histories whose delays differ within the cell. This is the same condition violated at a representational rather than a geometric level, and it is the subject of §11.

Table 1 summarises the reading. The three rows are one criterion, and the engineering response in each case is determined by which equivalence class the observable requires.

E.3 Branch preservation as a bit budget

Theorem 11 was stated for J equal branches carrying a uniform delay of width W. The argument generalises, and in doing so makes the cost comparable across the paper. Suppose branch j has probability p_j, and that at high rate the squared-error distortion of a scalar quantiser with K_j cells on branch j is $C_j K_j^{-2}$ (for a uniform width-W branch, $C_j = W^2/12$).

Proposition 21 (General branch-preservation cost)**.** *Under a total cell budget $\sum_j K_j = K$, the optimal branch-preserving allocation is $K_j \propto (p_j C_j)^{1/3}$, giving*

$$D^\star_{\mathrm{branch}}(K) = \frac{\left(\sum_{j=1}^{J} (p_j C_j)^{1/3}\right)^3}{K^2}, \tag{66}$$

while an unconstrained quantiser that may pool branches attains $D_{\text{pooled}}(K) \approx C_*/K^2$ *with* C_* *the pooled constant. [evidence type A]*

Proof. Minimise $\sum_j p_j C_j K_j^{-2}$ subject to $\sum_j K_j = K$. The stationarity condition is $-2p_jC_jK_j^{-3} = \lambda$ for all j, hence $K_j \propto (p_jC_j)^{1/3}$. Substituting $K_j = K(p_jC_j)^{1/3}/S$ with $S = \sum_i (p_iC_i)^{1/3}$ gives $D^\star = S^2K^{-2}\sum_j (p_jC_j)^{1/3} = S^3/K^2$. □

Corollary 22 (Theorem 11 is the symmetric instance)**.** *If the branches are equiprobable and share a common constant* C*, then* $S = J^{2/3}C^{1/3}$ *and* $D^\star_{\text{branch}}/D_{\text{pooled}} = J^2$*.*

The qualification matters and we state it plainly: this is a high-rate result for squared-error scalar quantisation of overlapping branch distributions under a shared total cell budget. It is not a universal penalty for preserving branch identity, and it does not apply when the branch-conditioned distributions are disjoint, in which case pooling is already lossless.

There is a second reading of J^2 that unifies it with the next subsection. Distinguishing J branches requires $\lceil \log_2 J \rceil$ bits of the address, so the effective cell count falls from K to K/J and the squared-error distortion rises by exactly $(K/(K/J))^2 = J^2 = 2^{2\log_2 J}$. The branch penalty of Theorem 11 is therefore not a separate phenomenon: it is the per-bit cost of identity, evaluated at $b = \log_2 J$ bits.

E.4 What a reserved history bit actually costs

Finally, reserve b bits of a fixed B-bit address for a history invariant—for example a lap counter with $b = \lceil \log_2(R+1) \rceil$ encoding up to R traversals. Then $K \le 2^{B-b}$, and the model of Theorem 12 reads

$$R(K) - \sigma^2 = \frac{A}{K^2} + \frac{A}{MK} + \frac{\nu^2 K}{M}, \qquad K \le 2^{B-b}. \tag{67}$$

It is tempting to conclude that each history bit multiplies distortion by four, since $K \mapsto K/2$ doubles the A/K^2 term. That conclusion is only sometimes right, and comparing the reserved bit against the calibration-optimal cell count $K_{\text{opt}} \approx (2AM/\nu^2)^{1/3}$ separates the cases.

Proposition 23 (Cost of history bits)**.** *Let* $K_{\text{opt}} = (2AM/\nu^2)^{1/3}$ *and suppose* K *is chosen optimally subject to* $K \le 2^{B-b}$*. Then:*

(i) if $K_{\text{opt}} \le 2^{B-b}$*, the address budget does not bind, the optimal cell count is* K_{opt}*, and the optimal error is* independent of b*: reserved history bits are free, because the calibration budget, not the address, is what limits resolution;*

(ii) if $K_{\text{opt}} > 2^{B-b}$*, the address binds and* $K = 2^{B-b}$*, whence*

$$R(b) - \sigma^2 = \frac{A\,2^{2b}}{2^{2B}} + \frac{A\,2^b}{2^B M} + \frac{\nu^2}{M}\,2^{B-b}; \tag{68}$$

the quantization term grows by exactly 2^{2b} *but the calibration term* decays *by* 2^{-b}*, so the net penalty is strictly below* 2^{2b} *and is negative for sufficiently large* ν^2*.*

[evidence type A within the stated model]

Proof. Immediate from Theorem 12 and $K \le 2^{B-b}$. In (ii) the third term of R decreases in b, so $R(b) - R(0) < (2^{2b}-1)A/2^{2B}$ plus a 2^b-order correction; and $\partial_b R$ at $b = 0$ is $\ln 2\,(2A/2^{2B} + A/(2^B M) - \nu^2 2^B/M)$, which is negative whenever the per-cell calibration noise dominates. □

Part (i) is the practically important one, and it explains an apparent tension in the literature on the prototype: adding a lap counter is often nearly free, because the binding constraint was never the address width. Part (ii) also shows that coarsening a partition can *reduce* total measurement error when calibration is noisy, which is the same effect as the constant-delay counterexample of §14. The two are the same mechanism seen from opposite directions: cells cost calibration samples.

E.5 Statement of the framework, and an open problem

The results of this paper can now be stated as one thesis. A scientifically valid measurement representation is an encoding of histories whose fibres respect the required observable equivalence classes, subject to finite hardware and calibration resources:

$$\begin{aligned} &\min_{E,\,g,\,K,\,b,\,\{m_k\}} \quad \mathbb{E}\big[d(Y, g(E(H)))\big] + \sum_k \frac{p_k v_k}{m_k} \\ &\text{subject to} \quad E(H_1) = E(H_2) \Rightarrow Y(H_1) = Y(H_2), \quad \log_2 K + b \le B, \quad \sum_k m_k \le M, \end{aligned} \tag{69}$$

equivalently by Theorem 20: choose the coarsest encoding whose fibres lie inside the level sets of Y, then spend the remaining address and calibration budget optimally. On this reading, topology, branch identity, address capacity, calibration budget and observable distortion are not independent design knobs but coupled resources, which is the claim of §13 made precise.

What is missing is a single lower bound coupling all of them. The components are individually known—J^2 for identity (Corollary 22), 2^{2b} for reserved history bits in the address-limited regime (Proposition 23), and the $M^{-2/3}$ law of Theorem 12—but their *joint* optimum, in which J, R, B and M trade against one another, has not been derived here. Proposition 23 indicates that the answer is not a product of the individual factors, since the calibration term enters with the opposite sign. Obtaining that bound, and identifying which of the four resources is binding for a given instrument class, is the natural quantitative continuation of this work.

Data and code availability

The released artifacts accompany this manuscript: the synthetic ray generator and deterministic split rule (`code/generate.py`), the uniform and chart-legal DP partition code with 30-seed evaluation (`code/partition.py`), the per-seed results (`code/results_exp5_seeds.csv`), the budget-scaling, control, analytic-partition and chart-legality analysis behind Tables in §12 (`code/scaling.py`, `code/results_exp5_scaling.csv`), the Ada/SPARK encoder with its exhaustive runtime test (`code/chart_encoder.*`, `code/test_encoder.adb`), and the GNATprove transcript (`code/gnatprove-transcrip`
The two calibration constants of the reference generator are documented in its source; the chart table is the immutable table of Experiment 4. An independent audit of these artifacts, including the corrections it prompted, is released as `code/AUDIT.md`.

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